

SINGULARITIES IN COMMUTATIVE ALGEBRA

IZ TECÚM

ABSTRACT. SLMath notes.

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¹Based on lectures by Eloísa Grifo, Benjamin Briggs, and Josh Pollitz [4] .

Part 1. Lecture A²

1. RESOLUTIONS AND REGULARITY

This section relates minimal free resolutions, regular local rings, and Koszul homology. Throughout, assume [1, 6, 8]

$(R, \mathfrak{m}, k)$ commutative noetherian local, $\mathfrak{m}$ = the maximal ideal of R , $k := R/\mathfrak{m}$.
 $\text{mod}(R) := \{N : N \text{ finitely generated over } R\}$, $M \in \text{mod}(R)$.

1.1. Minimal Resolutions.

Definition 1.1.1 (Minimal generators). For $M \in \text{mod}(R)$, set

$$\mu_R(M) := \dim_k(M/\mathfrak{m}M) \stackrel{[10, \S 21]}{\iff} \mu_R(M) = \min \{n \geq 0 : M = Rm_1 + \cdots + Rm_n\}.$$

Example 1.1.2 (Standard presentation). For a field k , set

$$Q := k[[x_1, \dots, x_e]], \quad \mathfrak{n} := (x_1, \dots, x_e), \quad I \subseteq \mathfrak{n}^2, \quad R := Q/I, \quad \mathfrak{m} := \mathfrak{n}/I.$$

$$R/\mathfrak{m} \cong k, \quad \mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{n}/(\mathfrak{n}^2 + I) = \mathfrak{n}/\mathfrak{n}^2 \cong k^e \stackrel{1.1.1}{\implies} \mu_R(\mathfrak{m}) = e.$$

Definition 1.1.3 (Free resolution). For $M \in \text{mod}(R)$, let

$$F = (\cdots \xrightarrow{\partial_3} F_2 \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \longrightarrow 0), \quad F_i \cong R^{r_i}, \quad r_i < \infty.$$

$$F \text{ a free resolution of } M \iff \cdots \longrightarrow F_2 \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0 \text{ is exact.}$$

$$F \text{ a free resolution of } M \stackrel{1.1.3}{\iff} H_0(F) \cong M, \quad H_i(F) = 0 \quad (i \geq 1).$$

$$F \text{ minimal} \iff \partial_i(F_i) \subseteq \mathfrak{m}F_{i-1} \quad (i \geq 1), \quad \partial(F) \subseteq \mathfrak{m}F.$$

Definition 1.1.4 (Successive syzygies). For $M \in \text{mod}(R)$, choose [1, §1]

$$\Omega_0^R(M) := M, \quad F_i := R^{\mu_R(\Omega_i^R(M))}, \quad \pi_i : F_i \twoheadrightarrow \Omega_i^R(M) \text{ minimal} \quad (i \geq 0).$$

For $i \geq 0$, set

$$\Omega_{i+1}^R(M) := \ker(\pi_i), \quad \ker(\pi_i) \subseteq \mathfrak{m}F_i, \quad \varepsilon := \pi_0,$$

$$\iota_i : \Omega_i^R(M) \hookrightarrow F_{i-1}, \quad \partial_i := \iota_i \circ \pi_i \quad (i \geq 1).$$

$$\partial_i(F_i) \subseteq \mathfrak{m}F_{i-1} \stackrel{1.1.3}{\implies} F \xrightarrow{\varepsilon} M \text{ is a minimal free resolution.}$$

The recursive factorizations assemble into the following exact diagram.

$$\begin{array}{ccccccc} \cdots & \longrightarrow & F_3 & \xrightarrow{\partial_3} & F_2 & \xrightarrow{\partial_2} & F_1 & \xrightarrow{\partial_1} & F_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0 \\ & & \searrow \pi_3 & & \nearrow \iota_3 & \searrow \pi_2 & & \nearrow \iota_2 & \searrow \pi_1 & & \nearrow \iota_1 & & \\ & & & & \Omega_3^R(M) & & \Omega_2^R(M) & & \Omega_1^R(M) & & & & \end{array}$$

Theorem 1.1.5 (Existence and uniqueness). Every $M \in \text{mod}(R)$ admits a minimal free resolution.

For minimal free resolutions F, F' of M and every free resolution G of M ,

$$F \cong F', \quad G \cong F \oplus C, \quad C \text{ contractible.}$$

The comparison is represented by the following commutative diagram.

²Based on lectures by Eloísa Grifo, Benjamin Briggs, and Josh Pollitz [4].

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & F_2 & \xrightarrow{\partial_2} & F_1 & \xrightarrow{\partial_1} & F_0 & \xrightarrow{\varepsilon} & M \longrightarrow 0 \\
& & \downarrow \varphi_2 \cong & & \downarrow \varphi_1 \cong & & \downarrow \varphi_0 \cong & & \parallel \\
\cdots & \longrightarrow & F'_2 & \xrightarrow{\partial'_2} & F'_1 & \xrightarrow{\partial'_1} & F'_0 & \xrightarrow{\varepsilon'} & M \longrightarrow 0
\end{array}$$

Proof. Let F, F' be minimal resolutions of M , let G be any free resolution, and choose comparison maps $\varphi : F \rightarrow F'$ and $\psi : F' \rightarrow F$ lifting id_M ; minimality makes the comparison homotopies invertible modulo $\mathfrak{m}$ [1, 13, §1].

$M \in \text{mod}(R) \xrightarrow{1.1.4} M$ admits a minimal free resolution, F, F' minimal $\xrightarrow{1.1.3} \partial_F \otimes_R k = 0 = \partial_{F'} \otimes_R k$,

$$\varphi, \psi \text{ lift } \text{id}_M \xrightarrow{[1, 13, \S 1]} \psi \circ \varphi \simeq \text{id}_F, \quad \varphi \circ \psi \simeq \text{id}_{F'},$$

$$\partial_F \otimes_R k = 0 = \partial_{F'} \otimes_R k \xrightarrow{1.1.3} (\psi_i \otimes_R k) \circ (\varphi_i \otimes_R k) = \text{id}_{F_i \otimes_R k}, \quad (\varphi_i \otimes_R k) \circ (\psi_i \otimes_R k) = \text{id}_{F'_i \otimes_R k},$$

$$\varphi_i \otimes_R k \text{ invertible} \xrightarrow{[10, \S 2]} \varphi_i \text{ invertible} \quad (i \geq 0), \quad (\varphi_i \otimes_R k)^{-1} = \psi_i \otimes_R k,$$

$$G \text{ free resolution} \xrightarrow{[1, \S 1], [16]} G \cong F_{\min} \oplus C, \quad F_{\min} \cong F, \quad C \text{ contractible.}$$

Thus Nakayama's lemma upgrades the inverse comparison maps over k to a degreewise isomorphism over R , and every nonminimal cancellation remains in C . $\square$

Proposition 1.1.6 (Minimality criterion). *For a free resolution F of M ,*

$$F \text{ minimal} \iff \partial \otimes_R k = 0 \iff \text{every entry of every } \partial_i \text{ lies in } \mathfrak{m}.$$

For every $i \geq 0$,

$$F_i \twoheadrightarrow \Omega_i^R(M) \text{ minimal} \xLeftrightarrow{1.1.1} \text{rank}_R(F_i) = \mu_R(\Omega_i^R(M)).$$

Proof. Fix bases of F_i and write $A_i := [\partial_i]$; reduction modulo $\mathfrak{m}$ detects both minimality and the number of generators.

$$F \text{ minimal} \xLeftrightarrow{1.1.3} \partial_i(F_i) \subseteq \mathfrak{m}F_{i-1} \iff (F_i/\mathfrak{m}F_i \xrightarrow{\bar{\partial}_i} F_{i-1}/\mathfrak{m}F_{i-1}) = 0 \iff \partial_i \otimes_R k = 0 \quad (i \geq 1),$$

$$\partial_i \otimes_R k = 0 \iff \bar{A}_i = 0 \iff (A_i)_{pq} \in \mathfrak{m} \quad (p, q, i),$$

$$\ker(F_i \twoheadrightarrow \Omega_i^R(M)) = \partial_{i+1}(F_{i+1}) \subseteq \mathfrak{m}F_i \xLeftrightarrow{[1, \S 1], 1.1.1} \text{rank}_R(F_i) = \mu_R(\Omega_i^R(M)).$$

Thus vanishing after base change, matrix entries in $\mathfrak{m}$, and minimal covers of the successive syzygies are equivalent. $\square$

Definition 1.1.7 (Betti numbers). For the minimal free resolution F of M , set

$$\beta_i^R(M) := \text{rank}_R(F_i), \quad F_i \cong R^{\beta_i^R(M)}.$$

$$\text{pd}_R(0) := -\infty, \quad \text{pd}_R(M) := \sup\{i \geq 0 : F_i \neq 0\} \in \mathbb{N} \cup \{\infty\} \quad (M \neq 0).$$

By Theorem 1.1.5, these quantities are independent of F .

Proposition 1.1.8 (Betti numbers via Tor and Ext). *For every $i \geq 0$,*

$$\beta_i^R(M) = \dim_k \text{Tor}_i^R(M, k) = \dim_k \text{Ext}_R^i(M, k).$$

Minimal base change is represented by the following diagram.

$$\begin{array}{ccccc}
F_{i+1} & \xrightarrow{\partial_{i+1}} & F_i & \xrightarrow{\partial_i} & F_{i-1} \\
\downarrow & & \downarrow & & \downarrow \\
F_{i+1} \otimes_R k & \xrightarrow{0} & F_i \otimes_R k & \xrightarrow{0} & F_{i-1} \otimes_R k \\
& & \cong \downarrow & & \\
& & H_i(F \otimes_R k) = \mathrm{Tor}_i^R(M, k) \cong k^{\beta_i^R(M)} & &
\end{array}$$

Proof. Minimality annihilates the induced differentials over k , so both derived functors are read directly from the free ranks of F .

$$F \text{ minimal} \xrightarrow{1.1.6} \partial \otimes_R k = 0, \quad \mathrm{Hom}_R(\partial, k) = 0, \quad H_i(F \otimes_R k) = F_i \otimes_R k, \quad H^i(\mathrm{Hom}_R(F, k)) = \mathrm{Hom}_R(F_i, k),$$

$$\begin{aligned}
F \xrightarrow{\varepsilon} M \text{ free resolution} &\xrightarrow{[1, \S 1], [13, 16]} \mathrm{Tor}_i^R(M, k) = H_i(F \otimes_R k), & \mathrm{Ext}_R^i(M, k) &= H^i(\mathrm{Hom}_R(F, k)), \\
F_i \cong R^{\beta_i^R(M)} &\xrightarrow{1.1.7} F_i \otimes_R k \cong k^{\beta_i^R(M)}, & \mathrm{Hom}_R(F_i, k) &\cong k^{\beta_i^R(M)}, & \dim_k \mathrm{Tor}_i^R(M, k) &= \beta_i^R(M) = \dim_k \mathrm{Ext}_R^i(M, k).
\end{aligned}$$

Thus the common k -dimension is the minimal free rank $\beta_i^R(M)$. $\square$

Example 1.1.9 (Residue field). For $R = R_1 := k[[x]]$, the minimal resolution is the following exact sequence.

$$\begin{aligned}
0 &\rightarrow R \xrightarrow{x} R \xrightarrow{\varepsilon} k \rightarrow 0 \\
\beta_0^{R_1}(k) &= \beta_1^{R_1}(k) = 1, \quad \mathrm{pd}_{R_1}(k) = 1.
\end{aligned}$$

For $R = R_2 := k[[x]]/(x^2)$, the minimal resolution is the following periodic sequence.

$$\begin{aligned}
\cdots &\xrightarrow{x} R \xrightarrow{x} R \xrightarrow{x} R \xrightarrow{\varepsilon} k \rightarrow 0 \\
\ker(R_2 \xrightarrow{x} R_2) &= (x) = \mathrm{im}(R_2 \xrightarrow{x} R_2) \xrightarrow{1.1.3, 1.1.6, 1.1.7} \beta_i^{R_2}(k) = 1 \quad (i \geq 0), \quad \mathrm{pd}_{R_2}(k) = \infty.
\end{aligned}$$

1.2. Regularity.

Definition 1.2.1 (Regular local ring). For $(R, \mathfrak{m}, k)$ as above, set

$$\mathrm{embdim}(R) := \mu_R(\mathfrak{m}) \xleftarrow{1.1.1} \mathrm{embdim}(R) = \dim_k(\mathfrak{m}/\mathfrak{m}^2).$$

$$R \text{ regular} \iff \mathrm{embdim}(R) = \dim(R).$$

[Krull height theorem] For $(R, \mathfrak{m}, k)$ as above [10, §13], [6, §2.2],

$$\dim(R) = \mathrm{ht}(\mathfrak{m}) \leq \mu_R(\mathfrak{m}) = \mathrm{embdim}(R).$$

[Auslander–Buchsbaum–Serre] For $(R, \mathfrak{m}, k)$ as above [10, §19], [6, §2.2], [1, §1],

$$R \text{ regular} \iff \mathrm{pd}_R(k) < \infty \iff \mathrm{pd}_R(M) < \infty \quad \text{for every } M \in \mathrm{mod}(R),$$

$$R \text{ regular} \xrightarrow{[10, \S 19], [6, \S 2.2], [1, \S 1]} \mathrm{pd}_R(k) = \dim(R).$$

Corollary 1.2.2 (First singularity test).

$$k[[x]] \text{ regular}, \quad k[[x]]/(x^2) \text{ not regular.}$$

Proof. The two residue-field resolutions of Example 1.1.9 are distinguished by the finite-projective-dimension criterion in Fact 1.2.

$$\mathrm{pd}_{k[[x]]}(k) = 1 \xrightarrow{1.1.9, 1.2} k[[x]] \text{ regular}, \quad \mathrm{pd}_{k[[x]]/(x^2)}(k) = \infty \xrightarrow{1.1.9, 1.2} k[[x]]/(x^2) \text{ not regular.}$$

Thus finite projective dimension occurs for the power-series ring and fails for its singular quotient. $\square$

[Cohen structure theorem] For the $\mathfrak{m}$ -adic completion of R [10, §29], [6], set

$$\widehat{R} := \varprojlim_n R/\mathfrak{m}^n.$$

There exist

$$(Q, \mathfrak{n}, k) \text{ complete regular local, } I \subseteq \mathfrak{n}^2, \quad \widehat{R} \cong Q/I.$$

The completion and its Cohen presentation fit into the following commutative diagram.

$$\begin{array}{ccccccc} & & R & \xrightarrow{\iota} & \widehat{R} & & \\ & & \searrow & & \downarrow \cong & & \\ 0 & \longrightarrow & I & \hookrightarrow & Q & \xrightarrow{\pi} & Q/I \longrightarrow 0 \end{array}$$

Definition 1.2.3 (Complete intersection). For $\widehat{R} \cong Q/I$ as in Fact 1.2 [6, §2.3], set

$$Q \twoheadrightarrow \widehat{R} \cong Q/I, \quad I \subseteq \mathfrak{n}^2, \quad Q \twoheadrightarrow \widehat{R} \text{ a minimal Cohen presentation.}$$

R a complete intersection $\iff I = (f_1, \dots, f_c)$, $f_1, \dots, f_c$ a Q -regular sequence, $\text{codim}(R) := c$.

Proposition 1.2.4 (Regularity in a minimal presentation). For a minimal Cohen presentation $\widehat{R} \cong Q/I$,

$$R \text{ regular} \iff I = 0.$$

Proof. Completion preserves dimension and embedding dimension, while $I \subseteq \mathfrak{n}^2$ preserves the cotangent space of the minimal Cohen presentation. Set

$$\widehat{\mathfrak{m}} := \mathfrak{n}/I.$$

$$\begin{aligned} R \longrightarrow \widehat{R} &\xrightarrow{[10, \S 13]} \dim(R) = \dim(\widehat{R}), \quad \text{embdim}(R) = \text{embdim}(\widehat{R}), \quad R \text{ regular} \iff \widehat{R} \text{ regular}, \\ I \subseteq \mathfrak{n}^2 &\xrightarrow{1.2} \widehat{\mathfrak{m}}/\widehat{\mathfrak{m}}^2 \cong \mathfrak{n}/(\mathfrak{n}^2 + I) = \mathfrak{n}/\mathfrak{n}^2, \quad \text{embdim}(\widehat{R}) = \text{embdim}(Q) = \dim(Q), \\ R \text{ regular} &\xrightarrow{[10, \S 13]} \widehat{R} \text{ regular} \xrightarrow{1.2.1} \dim(\widehat{R}) = \dim(Q) \xrightarrow{1.2} \dim(Q/I) = \dim(Q), \\ \dim(Q/I) &= \dim(Q) \xrightarrow{[10, \S 14]} \text{ht}(I) = 0 \xrightarrow{[10, \S 14]} I = 0. \end{aligned}$$

Since Q is a domain, height zero forces $I = 0$, while $I = 0$ gives $\widehat{R} \cong Q$. $\square$

Theorem 1.2.5 (Conormal criterion). Let Q be regular local, $I \subsetneq Q$, and $R := Q/I$. Then

$$I \text{ generated by a } Q\text{-regular sequence} \iff \text{pd}_R(I/I^2) < \infty.$$

For $c \geq 0$, $I = (f_1, \dots, f_c)$, and a Q -regular sequence $f_1, \dots, f_c$, the conormal basis is the following isomorphism.

$$R^c = \bigoplus_{i=1}^c R e_i \xrightarrow[\cong]{e_i \mapsto f_i + I^2} I/I^2 = \bigoplus_{i=1}^c R(f_i + I^2)$$

Proof. The regular-sequence direction computes a free conormal module, while the converse applies the finite-projective-dimension theorem of Briggs.

$$\begin{aligned} f_1, \dots, f_c \text{ a } Q\text{-regular sequence} &\xrightarrow{[6, \S 2.3]} I/I^2 \cong R^c \xrightarrow{1.1.7} \text{pd}_R(I/I^2) \in \{-\infty, 0\}, \quad \text{pd}_R(I/I^2) < \infty, \\ \begin{cases} I = 0 & \xrightarrow{1.3.1} I \text{ generated by the empty } Q\text{-regular sequence,} \\ I \neq 0, \text{pd}_R(I/I^2) < \infty & \xrightarrow{[5]} I \text{ generated by a } Q\text{-regular sequence.} \end{cases} \end{aligned}$$

Thus the equivalence is the regular-ambient case of Vasconcelos' conjecture [5]. $\square$

1.3. Koszul Complexes.

Definition 1.3.1 (Regular sequence). For $M \in \text{mod}(R)$, $M \neq 0$, and $f \in R$ [6, §1.1],

$$f \text{ regular on } M \iff fM \neq M, \quad \ker(M \xrightarrow{f} M) = 0.$$

For $n = 0$, set

$$\mathbf{f} := \emptyset, \quad M/(\mathbf{f})M := M, \quad \mathbf{f} \text{ regular on } M.$$

For $n \geq 1$ and $\mathbf{f} := (f_1, \dots, f_n) \in \mathfrak{m}^n$, set

$$M_0 := M, \quad M_i := M/(f_1, \dots, f_i)M \quad (1 \leq i \leq n).$$

$$\mathbf{f} \text{ regular on } M \iff M_n \neq 0, \quad f_i M_{i-1} \neq M_{i-1}, \quad \ker(M_{i-1} \xrightarrow{f_i} M_{i-1}) = 0 \quad (1 \leq i \leq n).$$

[Regular parameters] For a minimal generating set $x_1, \dots, x_e$ of $\mathfrak{m}$ [6, §2.2], [10, §14],

$$R \text{ regular} \iff x_1, \dots, x_e \text{ an } R\text{-regular sequence.}$$

Definition 1.3.2 (Exterior algebra). For an R -module E , set

$$T_R(E) := \bigoplus_{i \geq 0} E^{\otimes i}, \quad \bigwedge_R E := T_R(E)/\langle e \otimes e : e \in E \rangle, \quad \bigwedge_R E = \bigoplus_{i \geq 0} \bigwedge_R^i E.$$

For homogeneous $a, b \in \bigwedge_R E$,

$$a \wedge b = (-1)^{|a||b|} b \wedge a, \quad |a| \text{ odd} \xrightarrow{1.3.2} a \wedge a = 0.$$

[Exterior basis] For $E = R^n$ with basis $e_1, \dots, e_n$ [6, §1.6], [16, §4.5],

$$\bigwedge_R^d E = \bigoplus_{1 \leq i_1 < \dots < i_d \leq n} R(e_{i_1} \wedge \dots \wedge e_{i_d}), \quad \text{rank}_R\left(\bigwedge_R^d E\right) = \binom{n}{d}.$$

Definition 1.3.3 (Koszul complex). For $\mathbf{f} := (f_1, \dots, f_n) \in R^n$ and $E := R^n$ with basis $e_1, \dots, e_n$, set

$$K^R(\mathbf{f})_d := \bigwedge_R^d E, \quad \partial(e_i) := f_i, \quad a, b \text{ homogeneous}, \quad \partial(a \wedge b) := \partial(a) \wedge b + (-1)^{|a|} a \wedge \partial(b).$$

$$\partial(e_{i_1} \wedge \dots \wedge e_{i_d}) := \sum_{j=1}^d (-1)^{j-1} f_{i_j} \cdot e_{i_1} \wedge \dots \wedge \widehat{e_{i_j}} \wedge \dots \wedge e_{i_d}.$$

These maps assemble into the following augmented complex.

$$0 \rightarrow K_n^R(\mathbf{f}) = \bigwedge_R^n R^n \xrightarrow{\partial_n} K_{n-1}^R(\mathbf{f}) = \bigwedge_R^{n-1} R^n \xrightarrow{\partial_{n-1}} \dots \rightarrow K_2^R(\mathbf{f}) = \bigwedge_R^2 R^n \xrightarrow{\partial_2} K_1^R(\mathbf{f}) = R^n \xrightarrow{\partial_1} K_0^R(\mathbf{f}) = R \xrightarrow{\varepsilon} R/(f_1, \dots, f_n) \rightarrow 0$$

Proposition 1.3.4 (Square zero). For the differential of Definition 1.3.3,

$$\partial^2 = 0.$$

Proof. Fix $u := e_{i_1} \wedge \dots \wedge e_{i_d}$; deleting e_{i_p} and e_{i_q} in opposite orders produces the same wedge with opposite signs. For $1 \leq p < q \leq d$, set

$$\begin{aligned} u_{pq} &:= e_{i_1} \wedge \dots \wedge \widehat{e_{i_p}} \wedge \dots \wedge \widehat{e_{i_q}} \wedge \dots \wedge e_{i_d}, \\ c_{pq}^{p,q} &:= (-1)^{p-1} (-1)^{q-2} f_{i_p} f_{i_q} = (-1)^{p+q-3} f_{i_p} f_{i_q}, \quad c_{pq}^{q,p} := (-1)^{q-1} (-1)^{p-1} f_{i_q} f_{i_p} = (-1)^{p+q-2} f_{i_p} f_{i_q}, \\ c_{pq}^{p,q} + c_{pq}^{q,p} &= ((-1)^{p+q-3} + (-1)^{p+q-2}) f_{i_p} f_{i_q} = 0, \\ \partial^2(u) &\xrightarrow{1.3.3} \sum_{1 \leq p < q \leq d} (c_{pq}^{p,q} + c_{pq}^{q,p}) \cdot u_{pq} = 0. \end{aligned}$$

Thus every unordered pair cancels, and the differential squares to zero. $\square$

Example 1.3.5 (Small Koszul complexes). For $f \in R$,

$$K^R(f) : 0 \longrightarrow Re_1 \xrightarrow{f} R \longrightarrow 0.$$

For $f_1, f_2 \in R$, the two-element Koszul complex is the following sequence.

$$\begin{aligned} 0 \longrightarrow R(e_1 \wedge e_2) \xrightarrow{\partial_2 = \begin{bmatrix} -f_2 \\ f_1 \end{bmatrix}} Re_1 \oplus Re_2 \xrightarrow{\partial_1 = (f_1 \ f_2)} R \xrightarrow{\varepsilon} R/(f_1, f_2) \longrightarrow 0 \\ \partial_1 \circ \partial_2 \xrightarrow{1.3.3} (f_1 \ f_2) \cdot \begin{bmatrix} -f_2 \\ f_1 \end{bmatrix} = 0. \end{aligned}$$

Proposition 1.3.6 (Degree-zero homology). For every sequence $\mathbf{f} = (f_1, \dots, f_n)$,

$$H_0(K^R(\mathbf{f})) = R/(f_1, \dots, f_n).$$

Proof. Degree zero has no outgoing differential, and the image of ∂_1 is the ideal generated by $\mathbf{f}$.

$$\begin{aligned} K^R(\mathbf{f})_0 = R, \quad K^R(\mathbf{f})_{-1} = 0 \xrightarrow{1.3.3} H_0(K^R(\mathbf{f})) = \text{coker}(R^n \xrightarrow{[f_1 \ \dots \ f_n]} R), \\ \text{im}(R^n \xrightarrow{[f_1 \ \dots \ f_n]} R) = (f_1, \dots, f_n) \xrightarrow{1.3.3} \text{coker}(R^n \xrightarrow{[f_1 \ \dots \ f_n]} R) = R/(f_1, \dots, f_n) \xrightarrow{1.3.3} H_0(K^R(\mathbf{f})) = R/(f_1, \dots, f_n). \end{aligned}$$

Thus the degree-zero homology is the quotient by $(\mathbf{f})$. $\square$

Theorem 1.3.7 (Koszul criterion). For $\mathbf{f} = (f_1, \dots, f_n) \in \mathfrak{m}^n$,

$$\mathbf{f} \text{ an } R\text{-regular sequence} \iff K^R(\mathbf{f}) \text{ a free resolution of } R/(\mathbf{f}) \iff H_1(K^R(\mathbf{f})) = 0.$$

Proof. Koszul homology rigidity reduces acyclicity to degree one, and degree-one vanishing detects regularity.

$$\begin{aligned} H_1(K^R(\mathbf{f})) = 0 \xrightarrow{[6, \S 1.6]} H_i(K^R(\mathbf{f})) = 0 \quad (i \geq 1) \xrightarrow{1.1.3, 1.3.6} K^R(\mathbf{f}) \text{ a free resolution of } R/(\mathbf{f}), \\ \mathbf{f} \text{ an } R\text{-regular sequence} \xrightarrow{[6, \S 1.6], [8]} H_1(K^R(\mathbf{f})) = 0. \end{aligned}$$

Hence regularity, degree-one vanishing, and exactness of the augmented Koszul complex are equivalent. $\square$

Corollary 1.3.8 (Betti numbers of a quotient). For an R -regular sequence $\mathbf{f} = (f_1, \dots, f_c)$,

$$\beta_i^R(R/(\mathbf{f})) = \binom{c}{i} \quad (0 \leq i \leq c), \quad \text{pd}_R(R/(\mathbf{f})) = c.$$

Proof. For a regular sequence, the Koszul complex is the minimal resolution and its free ranks are the exterior ranks.

$$\mathbf{f} \text{ regular} \xrightarrow{1.3.7} K^R(\mathbf{f}) \text{ a free resolution of } R/(\mathbf{f}), \quad \mathbf{f} \in \mathfrak{m}^c \xrightarrow{1.3.3, 1.1.6} K^R(\mathbf{f}) \text{ minimal},$$

$$K^R(\mathbf{f})_i = \bigwedge_R^i R^c \xrightarrow{1.3, 1.1.7} \beta_i^R(R/(\mathbf{f})) = \binom{c}{i},$$

$$K^R(\mathbf{f})_i = 0 \quad (i > c), \quad K^R(\mathbf{f})_c = \bigwedge_R^c R^c \cong R \neq 0 \xrightarrow{1.1.7} \text{pd}_R(R/(\mathbf{f})) = c.$$

The nonzero top exterior power occurs in degree c , so the projective dimension is exactly c . $\square$

Corollary 1.3.9 (Koszul resolution of the residue field). For a minimal generating set $x_1, \dots, x_e$ of $\mathfrak{m}$,

$$R \text{ regular} \iff K^R(x_1, \dots, x_e) \text{ a free resolution of } k.$$

In this case,

$$\beta_i^R(k) = \binom{e}{i}, \quad \text{pd}_R(k) = e = \dim(R).$$

Proof. The minimal generators of $\mathfrak{m}$ form a regular sequence precisely when R is regular, so their Koszul complex controls the residue field.

R regular $\xLeftrightarrow{1.3} x_1, \dots, x_e$ an R -regular sequence, $x_1, \dots, x_e$ regular $\xLeftrightarrow{1.3.7} K^R(x_1, \dots, x_e)$ a free resolution of k ,

$$\beta_i^R(k) = \begin{pmatrix} e \\ i \end{pmatrix} \xrightarrow{1.3.8, 1.2} \text{pd}_R(k) = e = \dim(R).$$

Thus the exterior ranks give the Betti numbers, and Auslander–Buchsbaum–Serre identifies the terminal degree with $\dim(R)$. $\square$

2. DG RESOLUTIONS

DG resolutions retain multiplication while successively killing homology. Throughout, assume

$$(R, \mathfrak{m}, k) \text{ commutative Noetherian local, } A_i = 0 \quad (i < 0), \quad |\partial| = -1.$$

2.1. DG Algebras.

Definition 2.1.1 (Differential graded algebra). Let $A = \bigoplus_{i \geq 0} A_i$ be a graded R -algebra. A DG R -algebra structure on A is

$$\begin{aligned} R &\longrightarrow A_0, & r &\longmapsto r \cdot 1_A, & \partial(r \cdot 1_A) &= 0, & \partial_i : A_i &\longrightarrow A_{i-1}, & \partial_{i-1} \partial_i &= 0, & A_i \cdot A_j &\subseteq A_{i+j}, & 1_A &\in A_0, \\ a, b &\in A \text{ homogeneous,} & a \cdot b &= (-1)^{|a||b|} b \cdot a, & |a| \text{ odd} &\implies a^2 = 0, \\ & & \partial(a \cdot b) &= \partial(a) \cdot b + (-1)^{|a|} a \cdot \partial(b). \end{aligned}$$

For a graded A -module M , a DG A -module structure is

$$\begin{aligned} \partial_M : M_j &\longrightarrow M_{j-1}, & \partial_M^2 &= 0, & A_i \cdot M_j &\subseteq M_{i+j}, \\ \partial_M(a \cdot m) &= \partial(a) \cdot m + (-1)^{|a|} a \cdot \partial_M(m). \end{aligned}$$

For DG algebras A, B , set

$$\begin{aligned} \varphi : A &\longrightarrow B \text{ a DG-algebra map} \iff |\varphi| = 0, & \varphi(1_A) &= 1_B, & \varphi(a \cdot b) &= \varphi(a) \cdot \varphi(b), & \partial_B \varphi &= \varphi \partial_A, \\ \varphi &\text{ a quasi-isomorphism} \iff H_i(\varphi) : H_i(A) \xrightarrow{\cong} H_i(B) \quad (i \geq 0) \end{aligned}$$

[1, §2].

Proposition 2.1.2 (Homology algebra). For a DG algebra A , set

$$Z_i(A) := \ker(\partial_i), \quad B_i(A) := \text{im}(\partial_{i+1}), \quad Z(A) := \bigoplus_{i \geq 0} Z_i(A), \quad B(A) := \bigoplus_{i \geq 0} B_i(A).$$

Then

$$\begin{aligned} Z(A) &\subseteq A \text{ a graded subalgebra,} & B(A) &\subseteq Z(A) \text{ a graded ideal.} \\ H(A) &= Z(A)/B(A) \text{ a graded-commutative } H_0(A)\text{-algebra.} \end{aligned}$$

Proof. By Definition 2.1.1, the Leibniz rule sends products of cycles to cycles and products with boundaries to boundaries.

$$\begin{aligned} \partial(1_A) &= \partial(1_A \cdot 1_A) \xLeftrightarrow{2.1.1} \partial(1_A) = 2\partial(1_A) \implies \partial(1_A) = 0, & \partial(a) = \partial(b) = 0 &\xrightarrow{2.1.1} \partial(a \cdot b) = 0, \\ a &\in Z_p(A), & b = \partial(c) \in B_q(A) &\xrightarrow{2.1.1} \partial(b) = \partial^2(c) = 0, \\ a &\in Z_p(A), & b = \partial(c) \in B_q(A) &\xrightarrow{2.1.1} \partial(a \cdot c) = (-1)^p a \cdot b \implies a \cdot b = (-1)^p \partial(a \cdot c) \in B_{p+q}(A), \\ b &\in B_q(A) \subseteq Z_q(A), & a \cdot b &= (-1)^{pq} b \cdot a \implies B(A)Z(A) = Z(A)B(A) \subseteq B(A), \\ [a] \cdot [b] &:= [a \cdot b], & [a] \cdot [b] &= (-1)^{|a||b|} [b] \cdot [a], & |a| \text{ odd} &\implies [a]^2 = 0. \end{aligned}$$

Thus the quotient multiplication is well defined and has coefficient ring $H_0(A)$. $\square$

Example 2.1.3 (Koszul complex as a DG algebra). For $\mathbf{f} = (f_1, \dots, f_n) \in R^n$, set

$$A := K^R(\mathbf{f}) = \bigwedge_R(e_1, \dots, e_n), \quad |e_i| = 1, \quad \partial(e_i) = f_i.$$

By Definition 1.3.3, the differential is the unique extension satisfying Definition 2.1.1 [6, §1.6], [1, §2].

Definition 2.1.4 (Tensor product). For chain complexes A, B of R -modules and homogeneous $a \in A_i$, $b \in B_j$, set

$$(A \otimes_R B)_n := \bigoplus_{i+j=n} A_i \otimes_R B_j, \quad \partial(a \otimes b) := \partial(a) \otimes b + (-1)^{|a|} a \otimes \partial(b).$$

For DG R -algebras A, B , set

$$(a \otimes b) \cdot (a' \otimes b') := (-1)^{|b||a'|} (a \cdot a') \otimes (b \cdot b').$$

The signed total differential is represented by the following anticommuting grid [1, §2], [16, Chapter 2].

$$\begin{array}{ccccc} A_{i+1} \otimes_R B_{j+1} & \xrightarrow{\partial_A \otimes 1} & A_i \otimes_R B_{j+1} & \xrightarrow{\partial_A \otimes 1} & A_{i-1} \otimes_R B_{j+1} \\ \downarrow (-1)^{i+1} 1 \otimes \partial_B & & \downarrow (-1)^i 1 \otimes \partial_B & & \downarrow (-1)^{i-1} 1 \otimes \partial_B \\ A_{i+1} \otimes_R B_j & \xrightarrow{\partial_A \otimes 1} & A_i \otimes_R B_j & \xrightarrow{\partial_A \otimes 1} & A_{i-1} \otimes_R B_j \\ \downarrow (-1)^{i+1} 1 \otimes \partial_B & & \downarrow (-1)^i 1 \otimes \partial_B & & \downarrow (-1)^{i-1} 1 \otimes \partial_B \\ A_{i+1} \otimes_R B_{j-1} & \xrightarrow{\partial_A \otimes 1} & A_i \otimes_R B_{j-1} & \xrightarrow{\partial_A \otimes 1} & A_{i-1} \otimes_R B_{j-1} \end{array}$$

Proposition 2.1.5 (Tensor differential). *For the differential and product of Definition 2.1.4,*

$$\partial^2 = 0, \quad \partial(uv) = \partial(u)v + (-1)^{|u|} u \partial(v) \quad (u, v \text{ homogeneous}).$$

Proof. Fix homogeneous a, b, a', b' and set $p = |a|$, $q = |b|$, and $r = |a'|$; the cancellation in ∂^2 and the Koszul interchange sign control the calculation.

$$\partial^2(a \otimes b) \xrightarrow{2.1.4} \partial^2(a) \otimes b + (-1)^{p-1} \partial(a) \otimes \partial(b) + (-1)^p \partial(a) \otimes \partial(b) + a \otimes \partial^2(b) = 0,$$

$$\partial((a \otimes b) \cdot (a' \otimes b')) \xrightarrow{2.1.4} (-1)^{qr} \partial(a \cdot a') \otimes (b \cdot b') + (-1)^{qr+p+r} (a \cdot a') \otimes \partial(b \cdot b'),$$

$$L_A := (-1)^{qr} \left(\partial(a) \cdot a' \otimes (b \cdot b') + (-1)^p a \cdot \partial(a') \otimes (b \cdot b') \right),$$

$$L_B := (-1)^{qr+p+r} \left((a \cdot a') \otimes \partial(b) \cdot b' + (-1)^q (a \cdot a') \otimes b \cdot \partial(b') \right),$$

$$\partial((a \otimes b) \cdot (a' \otimes b')) = L_A + L_B,$$

$$\partial(a \otimes b) \cdot (a' \otimes b') = (-1)^{qr} \partial(a) \cdot a' \otimes (b \cdot b') + (-1)^{p+(q-1)r} (a \cdot a') \otimes \partial(b) \cdot b',$$

$$(-1)^{p+q} (a \otimes b) \cdot \partial(a' \otimes b') = (-1)^{p+qr} a \cdot \partial(a') \otimes (b \cdot b') + (-1)^{p+q+r+qr} (a \cdot a') \otimes b \cdot \partial(b').$$

Since $p + (q-1)r \equiv p + qr + r \pmod{2}$, the last two lines sum to the expanded differential of the product. $\square$

Corollary 2.1.6 (Koszul tensor decomposition). *For $\mathbf{f} = (f_1, \dots, f_n) \in R^n$,*

$$K^R(\mathbf{f}) \cong K^R(f_1) \otimes_R \cdots \otimes_R K^R(f_n) \quad \text{as DG } R\text{-algebras.}$$

On generators, the isomorphism is the upper edge of the following diagram.

$$\begin{array}{ccc}
K^R(f_1, \dots, f_n) & \xrightarrow[\cong]{\Phi(e_i)=1 \otimes \dots \otimes e_i \otimes \dots \otimes 1} & K^R(f_1) \otimes_R \dots \otimes_R K^R(f_n) \\
& \searrow \partial(e_i)=f_i & \swarrow \partial_\otimes (2.1.4) \\
& (\bigwedge_R(e_1, \dots, e_n), \partial(e_i)=f_i) &
\end{array}$$

Proof. The one-variable factors successively adjoin the exterior generators, and Definition 2.1.4 supplies exactly the Koszul signs.

$$\begin{aligned}
K^R(f) \otimes_R K^R(g) &\xrightarrow{2.1.4, 1.3.2} \bigwedge_R(e_f, e_g), \quad \partial(e_f) = f, \quad \partial(e_g) = g, \\
\bigwedge_R(e_f, e_g), \quad \partial(e_f) = f, \quad \partial(e_g) = g &\xrightarrow{1.3.3, 2.1.1} K^R(f, g), \\
K^R(f_1, \dots, f_{n-1}) \otimes_R K^R(f_n) &\cong K^R(f_1, \dots, f_n) \xrightarrow{[6, \S 1.6]} K^R(\mathbf{f}) \cong \bigotimes_{i=1}^n K^R(f_i).
\end{aligned}$$

Thus induction identifies both the product and the differential on every exterior monomial. $\square$

2.2. Tate Models.

Definition 2.2.1 (Adjoining a variable). Let A be a DG algebra and $z \in Z_{d-1}(A)$ homogeneous. Set

$$A[x \mid \partial(x) = z], \quad |x| = d, \quad \partial(x) = z,$$

with the differential extended by Definition 2.1.1. The underlying graded algebra is

$$d \text{ odd} \implies A[x] = A \oplus A \cdot x, \quad x^2 = 0.$$

For even d , set

$$\begin{aligned}
A[x] \text{ polynomial}, \quad \partial(x^j) &= j \cdot z \cdot x^{j-1}, \quad A\langle x \rangle \text{ divided-power}, \quad \partial(x^{(j)}) = z \cdot x^{(j-1)}, \\
x^{(p)} x^{(q)} &= \binom{p+q}{p} x^{(p+q)}.
\end{aligned}$$

Throughout, use the conventions of [15], [1, §6] and write

$A[X]$ for exterior odd and polynomial even variables, $A\langle X \rangle$ for exterior odd and divided-power even variables

Proposition 2.2.2 (Killed class). For $A[x \mid \partial(x) = z]$,

$$[z] = 0 \in H_{d-1}(A[x]).$$

If d is odd, there is a short exact sequence of complexes

$$(\Sigma^d A)_n := A_{n-d}, \quad \partial_{\Sigma^d A}(\Sigma^d a) := \Sigma^d \partial_A(a), \quad 0 \longrightarrow A \longrightarrow A[x] \longrightarrow \Sigma^d A \longrightarrow 0,$$

and its connecting map is multiplication by $[z]$ up to the suspension sign. The two maps occur in the following exact diagram.

$$0 \longrightarrow A \xrightarrow{\iota} A[x \mid \partial(x) = z] \xrightarrow{p} \Sigma^d A \longrightarrow 0$$

Furthermore,

$$H_{m-1}(A) \xleftarrow{\delta([a])=(-1)^{|a|}[az]} H_m(\Sigma^d A)$$

Proof. By Definition 2.2.1, the new generator bounds z , while the odd-degree decomposition identifies the quotient with a shift of A .

$$\begin{aligned} z = \partial(x) \xrightarrow{2.2.1} [z] = 0, \quad d \text{ odd} \xrightarrow{2.2.1} A[x] = A \oplus A \cdot x, \quad A[x]/A \cong \Sigma^d A, \\ \partial(a \cdot x) \xrightarrow{2.1.1} \partial(a) \cdot x + (-1)^{|a|} a \cdot \partial(x) \xrightarrow{2.2.1} \partial(a) \cdot x + (-1)^{|a|} a \cdot z, \\ \partial(a) = 0 \implies \partial(a \cdot x) = (-1)^{|a|} a \cdot z \implies \delta([a]) = (-1)^{|a|} [a \cdot z] = \pm [z] \cdot [a]. \end{aligned}$$

Hence the extension kills $[z]$, and the connecting morphism records precisely that multiplication. $\square$

Definition 2.2.3 (Even filtration). For an even polynomial variable x , set

$$A[x]_{\leq N} := \bigoplus_{j=0}^N A \cdot x^j, \quad A[x]_{\leq 0} \subseteq A[x]_{\leq 1} \subseteq \cdots, \quad A[x] = \bigcup_{N \geq 0} A[x]_{\leq N}.$$

Then

$$\partial(A[x]_{\leq N}) \subseteq A[x]_{\leq N}, \quad (A[x]_{\leq N}/A[x]_{\leq N-1})^{\natural} \cong (\Sigma^{Nd} A)^{\natural},$$

and every fixed homological degree stabilizes for $N \gg 0$ [1, §6].

Theorem 2.2.4 (Universal extension). *Let $\varphi : A \rightarrow B$ be a DG-algebra map and $Z \subseteq Z(A)$ homogeneous. For*

$$X := \{x_z : z \in Z\}, \quad |x_z| = |z| + 1, \quad \partial(x_z) = z,$$

there is a unique DG-algebra map

$$\tilde{\varphi} : A[X \mid \partial(x_z) = z] \longrightarrow B[X \mid \partial(x_z) = \varphi(z)], \quad \tilde{\varphi}|_A = \varphi, \quad \tilde{\varphi}(x_z) = x_z.$$

The extension is the unique dashed arrow in the following square.

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ A[X \mid \partial x = z] & \xrightarrow[\tilde{\varphi}]{} & B[X \mid \partial x = \varphi(z)] \end{array}$$

Moreover,

$$\varphi \text{ a quasi-isomorphism} \implies \tilde{\varphi} \text{ a quasi-isomorphism.}$$

Proof. The prescribed values force the graded-algebra map, and the boundary assignment makes it a chain map.

$$\begin{aligned} \partial(\tilde{\varphi}(x_z)) &= \partial(x_z) = \varphi(z) = \tilde{\varphi}(z) = \tilde{\varphi}(\partial(x_z)), \quad \tilde{\varphi}|_A = \varphi, \\ 0 \longrightarrow A &\longrightarrow A[x_z] \longrightarrow \Sigma^{|x_z|} A \longrightarrow 0, \quad 0 \longrightarrow B \longrightarrow B[x_z] \longrightarrow \Sigma^{|x_z|} B \longrightarrow 0, \\ |x_z| \text{ odd}, \quad H(\varphi) &\text{ bijective} \xrightarrow{2.2.2, [16, \S 1.3]} H(\tilde{\varphi}) \text{ bijective}, \\ |x_z| \text{ even}, \quad A[x_z]_{\leq N}/A[x_z]_{\leq N-1} &\cong \Sigma^{N|x_z|} A, \quad B[x_z]_{\leq N}/B[x_z]_{\leq N-1} \cong \Sigma^{N|x_z|} B, \\ H(\varphi) \text{ bijective} \xrightarrow{2.2.3, [16, \S 1.3]} H(A[x_z]_{\leq N}) &\xrightarrow{\cong} H(B[x_z]_{\leq N}) \quad (N \geq 0). \end{aligned}$$

Filtered colimits over N , then over finite subsets of X , preserve the quasi-isomorphism [1, §6]. $\square$

Corollary 2.2.5 (Prescribed boundaries). *Let $\varphi : A \rightarrow B$ be a DG-algebra map and $Z \subseteq Z(A)$ homogeneous. Assume*

$$b_z \in B_{|z|+1}, \quad \partial(b_z) = \varphi(z) \quad (z \in Z).$$

Then there is a unique DG-algebra map

$$\psi : A[X \mid \partial(x_z) = z] \longrightarrow B, \quad \psi|_A = \varphi, \quad \psi(x_z) = b_z.$$

Proof. The graded-algebra map is forced by the values on $A \cup X$, so only compatibility with the differential must be checked.

$$\begin{aligned} \partial(\psi(x_z)) &= \partial(b_z) = \varphi(z) = \psi(z) = \psi(\partial(x_z)), & \partial\psi(a) &= \partial\varphi(a) = \varphi\partial(a) = \psi\partial(a), \\ \partial\psi(u \cdot x_z) &\xrightarrow{2.1.1} \partial(\psi(u) \cdot b_z) = \partial\psi(u) \cdot b_z + (-1)^{|u|} \psi(u) \cdot \partial(b_z) = \psi(\partial(u \cdot x_z)). \end{aligned}$$

Thus Definition 2.1.1 extends the generator check to every monomial and gives uniqueness. $\square$

Definition 2.2.6 (Semifree extension). An extension $A \rightarrow A[X]$ is semifree when X admits a well-ordering [1, §2] satisfying

$$\partial(x) \in A[X_{<x}], \quad X_{<x} := \{y \in X : y < x\} \quad (x \in X)$$

Lemma 2.2.7 (Lifting lemma). *Let*

$\pi : C \twoheadrightarrow B$ a surjective quasi-isomorphism, $Q[X]$ semifree over Q , $f : Q[X] \rightarrow B$ a DG-algebra map.

If $f|_Q$ lifts to C , then the dashed map exists in

$$\begin{array}{ccc} Q[X] & \xrightarrow{f} & B \\ & \searrow \tilde{f} & \nearrow \pi \\ & C & \end{array}$$

and satisfies

$$\pi\tilde{f} = f.$$

The lift is unique up to Q -linear homotopy [1, §2].

Proof. Order X as in Definition 2.2.6 and assume the lift has been defined on $Q[X_{<x}]$; acyclicity of the kernel removes the obstruction at x .

$$\pi : C \twoheadrightarrow B \text{ a quasi-isomorphism} \xrightarrow{[16, \S 1.3]} 0 \rightarrow \ker(\pi) \rightarrow C \xrightarrow{\pi} B \rightarrow 0, \quad H(\ker(\pi)) = 0,$$

$$\begin{aligned} c_0 \in C_{|x|}, \quad \pi(c_0) &= f(x), \quad u := \partial(c_0) - \tilde{f}(\partial x) \in \ker(\pi)_{|x|-1}, \\ \partial(u) &= \partial^2(c_0) - \tilde{f}(\partial^2 x) = 0, \quad H(\ker(\pi)) = 0 \implies u = \partial(h), \quad h \in \ker(\pi)_{|x|}, \\ \tilde{f}(x) &:= c_0 - h, \quad \pi\tilde{f}(x) = f(x), \quad \partial\tilde{f}(x) = \partial(c_0) - \partial(h) = \tilde{f}(\partial x). \end{aligned}$$

The same well-ordered induction applied to two lifts constructs the relative Q -linear homotopy. $\square$

Definition 2.2.8 (Tate resolution). Let (Q, n, k) be Noetherian local, $I \subsetneq Q$, and $R = Q/I$. Choose

$$I = (f_1, \dots, f_n) \text{ minimally,} \quad X_1 := \{e_1, \dots, e_n\}, \quad |e_i| = 1, \quad \partial(e_i) = f_i,$$

and set

$$Q[X_1] := K^Q(f_1, \dots, f_n).$$

For $d \geq 2$, choose cycles $z_x \in Z_{d-1}(Q[X_{\leq d-1}])$ such that

$$\{[z_x] : x \in X_d\} \text{ minimally and finitely generates } H_{d-1}(Q[X_{\leq d-1}]) \text{ over } R, \quad |x| = d, \quad \partial(x) = z_x.$$

Set

$$X_{\leq d} := \bigcup_{1 \leq i \leq d} X_i, \quad X := \bigcup_{d \geq 1} X_d, \quad d \text{ odd} \implies X_d \text{ exterior}, \quad d \text{ even} \implies X_d \text{ polynomial}.$$

This semifree DG algebra is the Tate resolution $Q[X] \rightarrow R$ [15], [1, §6]. Its stages and killed homology are recorded by

$$Q \xrightarrow[|X_1|=1]{I} Q[X_1] = K^Q(\mathbf{f}) \xrightarrow[|X_2|=2]{H_1} Q[X_{\leq 2}] \longrightarrow \cdots \xrightarrow[|X_d|=d]{H_{d-1}} Q[X_{\leq d}]$$

$$\begin{array}{lll} H_0(Q[X_1]) \cong R & H_0(Q[X_{\leq 2}]) \cong R, & H_0(Q[X_{\leq d}]) \cong R, \\ & H_1(Q[X_{\leq 2}]) = 0 & H_i(Q[X_{\leq d}]) = 0 \quad (1 \leq i < d) \end{array}$$

Theorem 2.2.9 (Acyclicity). *For the Tate resolution of Definition 2.2.8,*

$$H_i(Q[X]) \cong \begin{cases} R, & i = 0, \\ 0, & i \geq 1, \end{cases} \quad Q[X] \xrightarrow{\sim} R.$$

Hence the underlying complex of $Q[X]$ is a free Q -resolution of R .

Proof. The variables in degree $i+1$ kill H_i , and variables of larger degree change neither chains of degree i nor boundaries entering that degree.

$$\begin{aligned} Q[X]_i &= Q[X_{\leq i}]_i, & Q[X]_{i+1} &= Q[X_{\leq i+1}]_{i+1}, & H_i(Q[X]) &= H_i(Q[X_{\leq i+1}]), \\ \{\partial(x) : x \in X_{i+1}\} &\text{ generates } H_i(Q[X_{\leq i}]) &\xrightarrow{2.2.8} H_i(Q[X_{\leq i+1}]) &= 0 \quad (i \geq 1), \\ H_0(Q[X_1]) &\xrightarrow{1.3.6} Q/(f_1, \dots, f_n) = Q/I = R, & Q[X]_0 &= Q[X_1]_0, & Q[X]_1 &= Q[X_1]_1, \\ H_0(Q[X]) &= R, & Q[X]^{\natural} &\text{ free over } Q &\xrightarrow{1.1.3} Q[X] &\xrightarrow{\sim} R \text{ a free resolution.} \end{aligned}$$

Thus the direct-limit DG algebra is acyclic in positive degrees and resolves R multiplicatively. $\square$

Corollary 2.2.10 (Stopping criterion). *For a minimal generating sequence $\mathbf{f} = (f_1, \dots, f_n)$ of I ,*

$$Q[X] = Q[X_1] = K^Q(\mathbf{f}) \iff \mathbf{f} \text{ a } Q\text{-regular sequence.}$$

Proof. The construction stops at stage one precisely when the first Koszul stage already has no positive homology.

$$\begin{aligned} Q[X] = Q[X_1] &\xrightarrow{2.2.8} H_1(K^Q(\mathbf{f})) = 0, & H_1(K^Q(\mathbf{f})) &= 0 \xrightarrow{1.3.7} \mathbf{f} \text{ a } Q\text{-regular sequence,} \\ \mathbf{f} \text{ regular} &\xrightarrow{1.3.7} H_i(K^Q(\mathbf{f})) = 0 \quad (i \geq 1) &\xrightarrow{2.2.8} X_d = \emptyset \quad (d \geq 2). \end{aligned}$$

Hence stage-one termination is exactly the complete-intersection case for the quotient map. $\square$

Definition 2.2.11 (Minimal model). Let $(Q, \mathfrak{n}, k)$ be local. A semifree resolution $Q[X] \xrightarrow{\sim} R$ is a minimal model if

$$(X) := Q[X] \cdot X, \quad \partial(X) \subseteq \mathfrak{n}Q[X] + (X)^2.$$

For the linear part, set

$$\partial^{\text{lin}}(x) \equiv \partial(x) \pmod{(X)^2} \quad (x \in X).$$

The variables grouped by homological degree form the following linear spine [1, §7].

$$\cdots \rightarrow \bigoplus_{x \in X_3} Qx \xrightarrow{\partial^{\text{lin}}} \bigoplus_{x \in X_2} Qx \xrightarrow{\partial^{\text{lin}}} \bigoplus_{x \in X_1} Qx \xrightarrow{\partial} Q \xrightarrow{\varepsilon} R \rightarrow 0$$

Proposition 2.2.12 (Tate models are minimal). *For the Tate resolution of Definition 2.2.8, assume*

$$I \subseteq \mathfrak{n}^2, \quad \{f_1, \dots, f_n\} \text{ minimally generates } I, \quad \{[\partial(x)] : x \in X_d\} \text{ minimally generates } H_{d-1}(Q[X_{< d}]).$$

Then $Q[X] \xrightarrow{\sim} R$ is a minimal model.

Proof. The initial boundaries lie in $\mathfrak{n}^2$, and at each later stage Nakayama forces every linear coefficient of a new boundary into $\mathfrak{n}$.

$$\begin{aligned} (\partial(X_1)) &= I \subseteq \mathfrak{n}^2 \subseteq \mathfrak{n}Q[X], & \partial(X_{\leq d}) &\subseteq \mathfrak{n}Q[X] + (X)^2, & y &\in X_{d+1}, \\ \partial(y) &= w + \sum_{x \in X_d} a_x x, & w &\in (X_{< d})^2, & a_x &\in Q, & \bar{a}_x &:= a_x + I \in R, \\ 0 = \partial^2(y) &= \partial(w) + \sum_{x \in X_d} a_x \partial(x) \implies \sum_{x \in X_d} \bar{a}_x [\partial(x)] = 0 & \text{ in } H_{d-1}(Q[X_{< d}]), \\ \{[\partial(x)] : x &\in X_d\} \text{ minimal} & \xrightarrow{[10, \S 2]} a_x &\in \mathfrak{n} \quad (x \in X_d), \\ \partial(y) &\in \mathfrak{n}Q[X] + (X)^2 \xrightarrow{2.2.11} \partial(X_{\leq d+1}) &\subseteq \mathfrak{n}Q[X] + (X)^2. \end{aligned}$$

Induction therefore gives the minimal-model containment for every Tate variable. $\square$

[Model versus resolution] Let Q be regular local and $R = Q/I$ not a complete intersection. Then

$$Q[X] \xrightarrow{\simeq} R \text{ a minimal model} \implies \#X = \infty, \quad \text{pd}_Q(R) < \infty.$$

Thus the underlying complex of a minimal model need not be the minimal free Q -resolution of R [1, §7]; the second assertion follows from Fact 1.2.

2.3. Deviations.

Definition 2.3.1 (Koszul complex of a ring). Choose a minimal generating set $\mathbf{x} = (x_1, \dots, x_e)$ of $\mathfrak{m}$, and set

$$e := \text{embdim}(R), \quad K^R := K^R(\mathbf{x}), \quad H(K^R) := \bigoplus_{i \geq 0} H_i(K^R).$$

By Proposition 2.1.2 and Proposition 1.3.6, one has [6, §1.6]

$$H_0(K^R) = R/(x_1, \dots, x_e) = R/\mathfrak{m} = k, \quad H(K^R) \text{ a graded-commutative } k\text{-algebra}.$$

Proposition 2.3.2 (Independence of generators). *For minimal generating sets $\mathbf{x} = (x_1, \dots, x_e)$ and $\mathbf{y} = (y_1, \dots, y_e)$ of $\mathfrak{m}$,*

$$K^R(\mathbf{x}) \cong K^R(\mathbf{y}) \quad \text{as DG } R\text{-algebras}.$$

Hence Definition 2.3.1 is independent of the chosen minimal generators [6, §1.6], [1, §2].

Proof. The two generating sets determine bases of $\mathfrak{m}/\mathfrak{m}^2$, so a change-of-basis matrix lifts to an invertible matrix over R .

$$\begin{aligned} \bar{\mathbf{x}}, \bar{\mathbf{y}} \text{ bases of the } k\text{-space } \mathfrak{m}/\mathfrak{m}^2 &\implies \mathbf{y} = U\mathbf{x}, & U &\in \text{M}_e(R), & \det(\bar{U}) &\neq 0, \\ \det(\bar{U}) \neq 0 &\implies \det(U) \in R^\times &\implies U &\in \text{GL}_e(R), \\ \Phi : K^R(\mathbf{y}) &\longrightarrow K^R(\mathbf{x}), & \Phi(e_{y_i}) &= \sum_{j=1}^e U_{ij} e_{x_j}, & \partial \Phi(e_{y_i}) &= \sum_{j=1}^e U_{ij} x_j = y_i, \\ \partial \Phi(e_{y_i}) &= \Phi \partial(e_{y_i}), & U &\in \text{GL}_e(R) \xrightarrow{1.3.2, 1.3.3} \Phi \text{ a DG-algebra isomorphism.} \end{aligned}$$

Thus every change of minimal generators induces an isomorphism of the corresponding Koszul DG algebras. $\square$

Example 2.3.3 (Three quotients). For

$$Q := k[[x, y]], \quad I_1 := (x^2 - y^2), \quad I_2 := (x^2, y^2), \quad I_3 := (x^2, xy),$$

one has

$$\begin{aligned} x^2 - y^2 & \text{ } Q\text{-regular}, & x^2, y^2 & \text{ a } Q\text{-regular sequence} \xrightarrow{1.2.3} Q/I_1, Q/I_2 \text{ complete intersections}, \\ \mu_Q(I_3) &= 2, & \sqrt{I_3} &= (x), & \text{ht}(I_3) &= 1 \xrightarrow{1.2, 1.2.3} Q/I_3 \text{ not a complete intersection.} \end{aligned}$$

[Completion invariants] Set

$$\widehat{R} := \varprojlim_n R/\mathfrak{m}^n, \quad \widehat{\mathfrak{m}} := \mathfrak{m}\widehat{R}, \quad \widehat{M} := \varprojlim_n M/\mathfrak{m}^n M \quad (M \text{ finitely generated}).$$

Then

$$\begin{aligned} R &\longrightarrow \widehat{R} \text{ flat and local,} & \mathfrak{m}^n/\mathfrak{m}^{n+1} &\cong \widehat{\mathfrak{m}}^n/\widehat{\mathfrak{m}}^{n+1}, & M \otimes_R \widehat{R} &\cong \widehat{M}, \\ \dim(R) &= \dim(\widehat{R}), & \text{depth}(R) &= \text{depth}(\widehat{R}), & \text{embdim}(R) &= \text{embdim}(\widehat{R}) \end{aligned}$$

[10, §8], [6, Chapter III].

Lemma 2.3.4 (Completion quasi-isomorphism). *Let M be a complex of finitely generated R -modules. Assume*

$$\ell_R(H_i(M)) < \infty \quad (i \geq 0).$$

Then the natural map

$$\eta_M : M \longrightarrow M \otimes_R \widehat{R}$$

is a quasi-isomorphism. Set

$$\eta_i : H_i(M) \longrightarrow H_i(M) \otimes_R \widehat{R}, \quad \beta_i : H_i(M) \otimes_R \widehat{R} \xrightarrow{\cong} H_i(M \otimes_R \widehat{R}).$$

These maps form the following square.

$$\begin{array}{ccc} H_i(M) & \xrightarrow[\cong]{H_i(\eta_M)} & H_i(M \otimes_R \widehat{R}) \\ \eta_i \downarrow \cong & & \cong \downarrow \beta_i^{-1} \\ H_i(M) \otimes_R \widehat{R} & \xlongequal{\quad} & H_i(M) \otimes_R \widehat{R} \end{array}$$

Proof. Flatness commutes completion with homology, and finite-length modules are already complete.

$$\begin{aligned} \widehat{R} \text{ flat over } R &\xrightarrow{2.3} H_i(M \otimes_R \widehat{R}) \cong H_i(M) \otimes_R \widehat{R}, \\ \ell_R(H_i(M)) < \infty &\implies \mathfrak{m}^{N_i} H_i(M) = 0 \quad (N_i \gg 0), \\ \mathfrak{m}^{N_i} H_i(M) = 0 &\xrightarrow{[10, \S 8]} H_i(M) \otimes_R \widehat{R} \cong H_i(M) \otimes_{R/\mathfrak{m}^{N_i}} \widehat{R}/\mathfrak{m}^{N_i} \widehat{R} \cong H_i(M), \\ H_i(\eta_M) = \beta_i \eta_i, &\quad \eta_i \text{ bijective,} \quad \beta_i \text{ bijective} \implies H_i(\eta_M) \text{ bijective.} \end{aligned}$$

Therefore completion preserves every homology group of M . □

Proposition 2.3.5 (Koszul complex and completion). *For $\mathbf{x} = (x_1, \dots, x_e)$ as in Definition 2.3.1,*

$$K^R(\mathbf{x}) \otimes_R \widehat{R} \cong K^{\widehat{R}}(x_1, \dots, x_e) \quad \text{as DG } \widehat{R}\text{-algebras,}$$

and

$$H(K^R) \cong H(K^{\widehat{R}}) \quad \text{as graded } k\text{-algebras.}$$

Set

$$\eta_K : K^R \longrightarrow K^R \otimes_R \widehat{R}, \quad a \longmapsto a \otimes 1, \quad \alpha : K^R \otimes_R \widehat{R} \xrightarrow{\cong} K^{\widehat{R}}.$$

The two identifications are the horizontal maps in

$$K^R \xrightarrow[\cong]{\eta_K : a \mapsto a \otimes 1} K^R \otimes_R \widehat{R} \xrightarrow[\cong]{\alpha} K^{\widehat{R}}$$

$$H(K^R) \xrightarrow[\cong]{H(\alpha \circ \eta_K)} H(K^{\widehat{R}})$$

Proof. The images of the minimal generators remain minimal after completion, and exterior multiplication gives an explicit null-homotopy for the maximal-ideal action on Koszul homology.

$$\begin{aligned} \mathfrak{m}/\mathfrak{m}^2 &\cong \widehat{\mathfrak{m}}/\widehat{\mathfrak{m}}^2 \xrightarrow{2.3} x_1, \dots, x_e \text{ minimally generate } \widehat{\mathfrak{m}}, \\ K^R(\mathbf{x}) \otimes_R \widehat{R} &= \bigwedge_{\widehat{R}} (e_1, \dots, e_e), \quad \partial(e_j) = x_j \xrightarrow{1.3.3} K^{\widehat{R}}(x_1, \dots, x_e), \\ h_j : K^R &\longrightarrow K^R, \quad h_j(a) := e_j \wedge a, \quad \partial h_j + h_j \partial = x_j \cdot 1_{K^R}, \\ (x_1, \dots, x_e) = \mathfrak{m} &\implies \mathfrak{m} H_i(K^R) = 0, \quad H_i(K^R) \text{ finitely generated} \implies \ell_R(H_i(K^R)) < \infty, \\ \ell_R(H_i(K^R)) < \infty &\xrightarrow{2.3.4} H(K^R) \xrightarrow{\cong} H(K^R \otimes_R \widehat{R}) \xrightarrow{\cong} H(K^{\widehat{R}}). \end{aligned}$$

Thus completion preserves both the Koszul DG algebra and its graded homology algebra. $\square$

Theorem 2.3.6 (First Koszul homology counts equations). *Let $(Q, \mathfrak{n}, k)$ be regular local and set*

$$R := Q/I, \quad I \subseteq \mathfrak{n}^2.$$

Then

$$\text{rank}_k H_1(K^R) = \mu_Q(I) \geq \text{ht}(I) = \text{embdim}(R) - \dim(R).$$

Proof. Choose a regular system of parameters of Q ; after base change its Koszul complex computes $\text{Tor}_1^Q(R, k)$, whose first group is the space of minimal equations of I .

$$\begin{aligned} \mathbf{x} = (x_1, \dots, x_e) &\text{ minimally generates } \mathfrak{n}, \quad e = \text{embdim}(Q) = \dim(Q), \quad I \subseteq \mathfrak{n}^2, \\ \mathfrak{m} = \mathfrak{n}/I, \quad \mathfrak{m}/\mathfrak{m}^2 &\cong \mathfrak{n}/(\mathfrak{n}^2 + I) = \mathfrak{n}/\mathfrak{n}^2 \implies \text{embdim}(R) = e, \\ K^R &\cong K^Q(\mathbf{x}) \otimes_Q R, \quad K^Q(\mathbf{x}) \xrightarrow{\cong} k \xrightarrow{1.3.9} H_1(K^R) \cong \text{Tor}_1^Q(k, R), \\ 0 \longrightarrow I &\longrightarrow Q \longrightarrow R \longrightarrow 0 \xrightarrow{[6, \S 1.6]} 0 \longrightarrow \text{Tor}_1^Q(k, R) \longrightarrow I/\mathfrak{n}I \longrightarrow k, \\ I \subseteq \mathfrak{n}^2 &\implies I/\mathfrak{n}I \longrightarrow k \text{ zero} \implies H_1(K^R) \cong I/\mathfrak{n}I \implies \text{rank}_k H_1(K^R) = \mu_Q(I), \\ \mu_Q(I) &\geq \text{ht}(I), \quad \text{ht}(I) = \dim(Q) - \dim(Q/I), \quad \dim(Q) = \text{embdim}(R) \xrightarrow{1.2, [10, \S 13]} \text{ht}(I) = \text{embdim}(R) - \dim(R). \end{aligned}$$

Hence degree-one Koszul homology counts the minimal defining equations and dominates their codimension [1, §7], [6, §1.6]. $\square$

Theorem 2.3.7 (Complete-intersection characterization). *Let*

$$\lambda_R : \bigwedge_k H_1(K^R) \longrightarrow H(K^R), \quad [z_1] \wedge \dots \wedge [z_q] \longmapsto [z_1 \dots z_q].$$

Then

$$R \text{ complete intersection} \iff \lambda_R \text{ an isomorphism} \iff \text{rank}_k H_1(K^R) = \text{embdim}(R) - \dim(R).$$

Proof. Completion preserves the Koszul algebra, and a minimal Cohen presentation converts the numerical condition into equality between the number and height of the defining equations.

$$\begin{aligned} H(K^R) &\cong H(K^{\widehat{R}}) \xrightarrow{2.3.5} \lambda_R \text{ an isomorphism} \iff \lambda_{\widehat{R}} \text{ an isomorphism}, \\ Q \twoheadrightarrow \widehat{R} = Q/I &\text{ a minimal Cohen presentation,} \quad Q \text{ complete regular local,} \quad I \subseteq \mathfrak{n}^2, \\ Q \twoheadrightarrow \widehat{R} = Q/I, \quad I &\subseteq \mathfrak{n}^2 \xrightarrow{2.3.6} \text{rank}_k H_1(K^R) = \mu_Q(I), \quad \text{embdim}(R) - \dim(R) = \text{ht}(I), \\ \text{rank}_k H_1(K^R) &= \text{embdim}(R) - \dim(R) \iff \mu_Q(I) = \text{ht}(I), \\ Q \text{ regular local,} \quad \mu_Q(I) &= \text{ht}(I) \xrightarrow{[6, \S 2.3]} I \text{ generated by a } Q\text{-regular sequence} \xrightarrow{1.2.3} R \text{ complete intersection,} \\ I \text{ generated by a regular sequence} &\xrightarrow{2.2.10} Q[X] = Q[X_1], \quad k[X_1] = \bigwedge_k (kX_1) \xrightarrow{[15], [1, \S 7]} \lambda_R \text{ an isomorphism,} \\ \lambda_R \text{ an isomorphism} &\xrightarrow{[15], [1, \S 7]} X_d = \emptyset \quad (d \geq 2) \xrightarrow{2.2.10} I \text{ generated by a regular sequence.} \end{aligned}$$

Thus the exterior-algebra structure and the equation count detect exactly the complete-intersection case. $\square$

Definition 2.3.8 (Acyclic closure). An acyclic closure of k over R is a semifree DG divided-power algebra

$$R\langle Y \rangle \xrightarrow{\simeq} k, \quad Y_n := \{y \in Y : |y| = n\},$$

built by minimally adjoining variables that kill the surviving homology. For $y \in Y_{2i}$, set

$$y^{(0)} = 1, \quad y^{(1)} = y, \quad y^{(p)}y^{(q)} = \binom{p+q}{p} y^{(p+q)}, \quad \partial(y^{(q)}) = \partial(y)y^{(q-1)}.$$

Odd variables are exterior, and acyclic closures are unique up to DG divided-power algebra isomorphism [9, §1], [1, §6].

Theorem 2.3.9 (Minimality of the acyclic closure). *Every local ring admits an acyclic closure satisfying*

$$\partial(Y) \subseteq \mathfrak{m}R\langle Y \rangle.$$

Its underlying complex is the minimal free resolution of k over R .

Proof. The Tate procedure chooses minimal homology generators at every stage, so all new boundaries lie in the maximal-ideal multiple of the preceding free algebra.

$$R\langle Y_{\leq n-1} \rangle, \quad \{[z_y] : y \in Y_n\} \text{ minimally generates } H_{n-1}(R\langle Y_{\leq n-1} \rangle), \quad \partial(y) = z_y,$$

$$\partial(Y_n) \subseteq \mathfrak{m}R\langle Y_{\leq n-1} \rangle \xrightarrow{[15], [9]} \partial(Y) \subseteq \mathfrak{m}R\langle Y \rangle,$$

$$H_i(R\langle Y \rangle) = 0 \quad (i \geq 1), \quad H_0(R\langle Y \rangle) = k \xrightarrow{[1, §6]} R\langle Y \rangle \xrightarrow{\simeq} k,$$

$$\partial(Y) \subseteq \mathfrak{m}R\langle Y \rangle \implies \partial(R\langle Y \rangle_i) \subseteq \mathfrak{m}R\langle Y \rangle_{i-1} \xrightarrow{1.1.6} R\langle Y \rangle \text{ minimal.}$$

Thus the multiplicative closure and the minimal free resolution have the same underlying complex. $\square$

Definition 2.3.10 (Deviations). For a minimal acyclic closure as in Theorem 2.3.9, set

$$\varepsilon_n(R) := \#Y_n \quad (n \geq 1).$$

These numbers are independent of the chosen closure [1, §7].

Theorem 2.3.11 (Deviations from a minimal model). *Fix a minimal Cohen presentation and a minimal model*

$$Q \twoheadrightarrow \widehat{R} = Q/I, \quad (Q, \mathfrak{n}, k) \text{ complete regular local}, \quad I \subseteq \mathfrak{n}^2, \quad Q[X] \xrightarrow{\simeq} \widehat{R}.$$

Then

$$\varepsilon_1(R) = \text{embdim}(R), \quad \varepsilon_{n+1}(R) = \#X_n \quad (n \geq 1).$$

After replacing R by $\widehat{R}$, write $R = Q/I$ and fix

$$\varepsilon_R : R \twoheadrightarrow k, \quad \varepsilon_Y : R\langle Y \rangle \xrightarrow{\simeq} k, \quad \varepsilon_X : Q[X] \xrightarrow{\simeq} R, \quad \varepsilon_{\text{fib}} : k[X] \longrightarrow k.$$

The acyclic closure and the closed fibre occupy the following diagram.

$$\begin{array}{ccc} R & \xrightarrow{\iota} & R\langle Y \rangle \\ & \searrow \varepsilon_R & \downarrow \simeq \varepsilon_Y \\ & & k \end{array} \quad \begin{array}{ccc} Q[X] & \xrightarrow[\simeq]{\varepsilon_X} & R \\ \downarrow k \otimes_Q - & & \downarrow \varepsilon_R \\ k[X] & \xrightarrow[\varepsilon_{\text{fib}}]{} & k \end{array}$$

For a regular system of parameters $\mathbf{x}$ of Q , the intrinsic Koszul complex and the derived fibre are compared by

$$K^R(\mathbf{x}) = K^Q(\mathbf{x}) \otimes_Q R \xleftarrow[1 \otimes \varepsilon_X]{\simeq} K^Q(\mathbf{x}) \otimes_Q Q[X] \xrightarrow[\simeq]{\varepsilon_R \otimes 1} k \otimes_Q Q[X] = k[X]$$

Proof. The comparison identifies the minimal-model indecomposable counts with the shifted acyclic-closure counts, while the first closure stage records the generators of $\mathfrak{m}$.

$$\begin{aligned} Y_1 &\longleftrightarrow \mathbf{x}_R, & \#Y_1 &= \mu_R(\mathfrak{m}) = \text{embdim}(R), \\ K^R(\mathbf{x}) &\xleftarrow{\simeq} K^Q(\mathbf{x}) \otimes_Q Q[X] \xrightarrow{\simeq} k \otimes_Q Q[X] = k[X], \\ Q[X] &\xrightarrow{\simeq} R, & R\langle Y \rangle &\xrightarrow{\simeq} k \xrightarrow{[1, \S 7], [9]} \#X_n = \varepsilon_{n+1}(R) = \#Y_{n+1} \quad (n \geq 1), \\ R &\longrightarrow \widehat{R} \xrightarrow{2.3, [1, \S 7]} \varepsilon_n(R) = \varepsilon_n(\widehat{R}) \quad (n \geq 1). \end{aligned}$$

Hence the degree- n variables of the minimal Cohen model count the $(n+1)$ -st deviations of R . $\square$

Corollary 2.3.12 (Deviation criteria). *For a local ring R ,*

$$R \text{ regular} \iff \varepsilon_n(R) = 0 \quad (n \geq 2), \quad R \text{ complete intersection} \iff \varepsilon_n(R) = 0 \quad (n \geq 3).$$

Proof. The regular case stops the acyclic closure after its degree-one Koszul stage, while the complete-intersection case stops the minimal Cohen model after degree one.

$$\begin{aligned} R \text{ regular} &\xleftrightarrow{1.3.9} K^R \xrightarrow{\simeq} k \xleftrightarrow{2.3.8} Y = Y_1 \xleftrightarrow{2.3.10} \varepsilon_n(R) = 0 \quad (n \geq 2), \\ R \text{ complete intersection} &\xleftrightarrow{2.3.7, 2.2.10} X = X_1 \xleftrightarrow{2.3.11} \varepsilon_n(R) = 0 \quad (n \geq 3), \\ \varepsilon_n(R) = 0 \quad (n \geq 3) &\xrightarrow{2.3.11} X_d = \emptyset \quad (d \geq 2) \xrightarrow{2.2.10} R \text{ complete intersection} \end{aligned}$$

. Thus deviation vanishing detects regularity one stage earlier than complete-intersection singularities [1, §7], [9]. $\square$

3. HOMOTOPY LIE THEORY

Homotopy Lie theory records the quadratic differential of a minimal model through derivations and Ext. Throughout, fix

$$\varphi : (Q, \mathfrak{n}, k) \twoheadrightarrow (R, \mathfrak{m}, k), \quad R = Q/I, \quad A = Q[X] \xrightarrow{\simeq} R \text{ minimal}, \quad \mathfrak{m}_A := \mathfrak{n}A + (X).$$

3.1. Ext and Differentials.

Definition 3.1.1 (Semiprojective modules and Ext). For a DG A -module P , set

$$P \text{ semiprojective} \iff \text{Hom}_A(P, -) \text{ preserves surjective quasi-isomorphisms.}$$

Then

$$P \text{ semifree} \xrightarrow{[1, \S 1]} P \text{ semiprojective}, \quad P \xrightarrow{\simeq} M \text{ semiprojective} \implies \text{Ext}_A^*(M, k) := H^*(\text{Hom}_A(P, k)).$$

Proposition 3.1.2 (Yoneda action). *For $f : M \rightarrow N$, the resolution-independent Yoneda action is contravariantly natural.*

$$\begin{array}{ccc} \text{Ext}_A^i(k, k) \otimes_k \text{Ext}_A^j(N, k) & \xrightarrow{\circ} & \text{Ext}_A^{i+j}(N, k) \\ \downarrow 1 \otimes f^* & & \downarrow f^* \\ \text{Ext}_A^i(k, k) \otimes_k \text{Ext}_A^j(M, k) & \xrightarrow{\circ} & \text{Ext}_A^{i+j}(M, k) \end{array}$$

Proof. Choose semiprojective resolutions P_k, P_M, P_N , representatives α, β , and a lift $\tilde{f} : P_M \rightarrow P_N$.

$$\begin{aligned} \alpha : P_k &\longrightarrow \Sigma^i P_k, & \beta : P_N &\longrightarrow \Sigma^j P_k, & [\alpha] \cdot [\beta] &:= [\Sigma^j \alpha \circ \beta], \\ \alpha &\simeq \alpha', & \beta &\simeq \beta' & \xrightarrow{2.2.7, [16, \S 1.3]} \Sigma^j \alpha \circ \beta &\simeq \Sigma^j \alpha' \circ \beta', \\ f^*([\beta]) &:= [\beta \circ \tilde{f}], & \tilde{f} &\simeq \tilde{f}' & \implies [\beta \circ \tilde{f}] &= [\beta \circ \tilde{f}'], \\ f^*([\alpha] \cdot [\beta]) &= [\Sigma^j \alpha \circ \beta \circ \tilde{f}] &= [\alpha] \cdot f^*([\beta]). \end{aligned}$$

Thus comparison maps and homotopy invariance give a well-defined natural action [1, §1, §10]. $\square$

Definition 3.1.3 (Connected covers). For $n \geq 1$, set

$$\begin{aligned} X_{<n} &:= \{x \in X : |x| < n\}, & X_{\geq n} &:= \{x \in X : |x| \geq n\}, & A_{<n} &:= Q[X_{<n}], \\ A^{(n)} &:= A \otimes_{A_{<n}} k = A/(n, X_{<n}) \cong k[X_{\geq n}], & A^{(1)} &= k[X] \simeq R \otimes_Q^L k \end{aligned}$$

[1, §7], [5].

$$A = Q[X] \xrightarrow{\text{mod } n} A^{(1)} = k[X] \xrightarrow{\text{mod } (X_1)} A^{(2)} = k[X_{\geq 2}] \xrightarrow{\text{mod } (X_2)} \dots \xrightarrow{\text{mod } (X_{n-1})} A^{(n)} = k[X_{\geq n}]$$

Proposition 3.1.4 (Models and complete intersections). For the minimal model $A = Q[X] \xrightarrow{\sim} R$,

A the minimal free Q -resolution of $R \iff I$ generated by a Q -regular sequence.

Proof. The construction stops at its Koszul stage exactly for a regular sequence.

$$I = (\mathbf{f}), \quad \mathbf{f} \text{ (Q)-regular} \xrightarrow{2.2.10} A = K^Q(\mathbf{f}) \xrightarrow{1.3.8} A \text{ the minimal free } Q\text{-resolution of } R,$$

$$A \text{ the minimal free } Q\text{-resolution of } R \xrightarrow{[1, \S 7]} X = X_1 \xrightarrow{2.2.10} I \text{ generated by a } Q\text{-regular sequence.}$$

Hence the multiplicative model is the minimal underlying resolution precisely in the complete-intersection case. $\square$

Example 3.1.5 (A finite DG algebra resolution). Let

$$Q := k[[x, y]], \quad I := (x^2, xy), \quad R := Q/I.$$

$$0 \longrightarrow Qb \xrightarrow{\partial_2 = \begin{bmatrix} -y \\ x \end{bmatrix}} Qe_1 \oplus Qe_2 \xrightarrow{\partial_1 = (x^2 \quad xy)} Q \xrightarrow{\varepsilon} Q/(x^2, xy) \longrightarrow 0$$

Set

$$|e_1| = |e_2| = 1, \quad |b| = 2, \quad \partial(e_1) = x^2, \quad \partial(e_2) = xy, \quad \partial(b) = -y \cdot e_1 + x \cdot e_2,$$

$$e_1 \cdot e_2 = x \cdot b, \quad e_1 \cdot b = e_2 \cdot b = b^2 = 0, \quad \partial(e_1 \cdot e_2 - x \cdot b) \xrightarrow{2.1.1} x^2 \cdot e_2 - xy \cdot e_1 - x \cdot (-y \cdot e_1 + x \cdot e_2) = 0.$$

Thus b realizes Corollary 2.2.5 for the cycle $-y \cdot e_1 + x \cdot e_2$.

Example 3.1.6 (The corresponding minimal model). For Q, I, R as in Example 3.1.5, set

$$X_1 := \{T_{1,1}, T_{1,2}\}, \quad \partial(T_{1,1}) = x^2, \quad \partial(T_{1,2}) = xy, \quad X_2 := \{T_{2,1}\}, \quad \partial(T_{2,1}) = -y \cdot T_{1,1} + x \cdot T_{1,2},$$

$$X_3 := \{T_{3,1}\}, \quad \partial(T_{3,1}) = -T_{1,1} \cdot T_{1,2} + x \cdot T_{2,1}, \quad X_4 := \{T_{4,1}, T_{4,2}\},$$

$$\partial(T_{4,1}) = -T_{1,2} \cdot T_{2,1} + y \cdot T_{3,1}, \quad \partial(T_{4,2}) = -T_{1,1} \cdot T_{2,1} + x \cdot T_{3,1}.$$

$$\begin{array}{c}
Q \\
\downarrow \begin{array}{l} \partial T_{1,1} = x^2 \\ \partial T_{1,2} = xy \end{array} \\
Q[T_{\leq 1}] \\
\downarrow \partial T_{2,1} = -yT_{1,1} + xT_{1,2} \\
Q[T_{\leq 2}] \\
\downarrow \partial T_{3,1} = -T_{1,1}T_{1,2} + xT_{2,1} \\
Q[T_{\leq 3}] \\
\downarrow \begin{array}{l} \partial T_{4,1} = -T_{1,2}T_{2,1} + yT_{3,1} \\ \partial T_{4,2} = -T_{1,1}T_{2,1} + xT_{3,1} \end{array} \\
Q[T_{\leq 4}] \\
\downarrow \\
\vdots
\end{array}$$

I not generated by a (Q) -regular sequence $\xRightarrow{2.3.3, 3.1.4} \#X = \infty$.

Definition 3.1.7 (DG derivation). For a DG Q -algebra B , a DG B -module M , and $r \in \mathbb{Z}$, set

$$\mathrm{Der}_Q^r(B, M) := \left\{ \theta \in \mathrm{Hom}_Q^r(B, M) : \theta(a \cdot b) = \theta(a) \cdot b + (-1)^{r|a|} a \cdot \theta(b) \right\},$$

$$\mathrm{Der}_Q(B, M) := \bigoplus_{r \in \mathbb{Z}} \mathrm{Der}_Q^r(B, M), \quad \partial(\theta) := \partial_M \theta - (-1)^{|\theta|} \theta \partial_B, \quad \theta \text{ a chain derivation} \iff \partial(\theta) = 0$$

[1, §7], [5].

Definition 3.1.8 (Universal derivation). For B as in Definition 3.1.7, let $d : B \rightarrow \Omega_{B/Q}$ satisfy

$$\mathrm{Hom}_B(\Omega_{B/Q}, M) \xrightarrow{\cong} \mathrm{Der}_Q(B, M), \quad f \longmapsto f \circ d$$

for every DG B -module M [5].

$$\begin{array}{ccc}
A & \xrightarrow{d} & \Omega_{A/Q} \\
& \searrow \theta & \downarrow \exists! \tilde{\theta} \\
& & M
\end{array}$$

Proposition 3.1.9 (Differentials of a semifree algebra). *For $A = Q[X]$,*

$$\Omega_{A/Q} \cong \bigoplus_{x \in X} A dx, \quad |dx| = |x|, \quad \partial(dx) = d(\partial x), \quad \text{Der}_Q^r(A, M) \cong \prod_{x \in X} M_{|x|+r}.$$

$$\cdots \xrightarrow{\partial} \bigoplus_{x \in X_3} A dx \xrightarrow{\partial} \bigoplus_{x \in X_2} A dx \xrightarrow{\partial} \bigoplus_{x \in X_1} A dx \longrightarrow 0$$

Proof. The free graded-algebra structure reduces a derivation to its values on X .

$$(m_x)_{x \in X} \in \prod_{x \in X} M_{|x|+r} \xrightarrow{3.1.7, [5]} \exists! \theta \in \text{Der}_Q^r(A, M), \quad \theta(x) = m_x,$$

$$\begin{aligned} \text{Hom}_A \left(\bigoplus_{x \in X} A dx, M \right) &\cong \prod_{x \in X} M_{|x|+*} \xleftrightarrow{3.1.8} \Omega_{A/Q} \cong \bigoplus_{x \in X} A dx, \\ d\partial_A &= \partial_\Omega d \implies \partial(dx) = d(\partial x). \end{aligned}$$

Hence the universal derivation identifies both the semifree basis and its differential. $\square$

Proposition 3.1.10 (Semifree flatness). *For a semifree DG A -module F ,*

$$M \xrightarrow{\sim} N \implies M \otimes_A F \xrightarrow{\sim} N \otimes_A F$$

[1, §1].

Theorem 3.1.11 (Differentials of a minimal model). *Set*

$$L_{R/Q} := R \otimes_A \Omega_{A/Q}.$$

Then

$$\Omega_{A/Q} \xrightarrow{\sim} L_{R/Q}, \quad L_{R/Q} \cong \bigoplus_{x \in X} R dx, \quad \partial(L_{R/Q}) \subseteq \mathfrak{m} L_{R/Q}, \quad H_1(L_{R/Q}) \cong I/I^2$$

[5].

$$\begin{array}{ccc} (\Omega_{A/Q} \otimes_A R)_2 & \xrightarrow{\partial_2} & (\Omega_{A/Q} \otimes_A R)_1 \\ \downarrow 0 & & \downarrow q(dy_j) = f_j + I^2 \\ 0 & \longrightarrow & I/I^2 \end{array}$$

$$\bigoplus_{i=1}^t R dz_i \xrightarrow{\partial_2 = (\bar{g}_{ij})} \bigoplus_{j=1}^n R dy_j \xrightarrow{dy_j \mapsto f_j + I^2} I/I^2 \longrightarrow 0$$

Proof. Choose the degree-one equations and degree-two syzygies of the minimal model.

$$I = (f_1, \dots, f_n), \quad X_1 = \{y_1, \dots, y_n\}, \quad \partial(y_j) = f_j, \quad X_2 = \{z_1, \dots, z_t\}, \quad \partial(z_i) = \sum_{j=1}^n g_{ij} \cdot y_j,$$

$$A \xrightarrow{\sim} R, \quad \Omega_{A/Q} \text{ semifree} \xrightarrow{3.1.9, 3.1.10} \Omega_{A/Q} \xrightarrow{\sim} R \otimes_A \Omega_{A/Q},$$

$$\partial(X) \subseteq \mathfrak{n}A + (X)^2 \xrightarrow{2.2.11, 3.1.9} \partial(L_{R/Q}) \subseteq \mathfrak{m} L_{R/Q},$$

$$\partial_2(dz_i) = \sum_{j=1}^n \bar{g}_{ij} \cdot dy_j, \quad \partial_1(dy_j) = 0, \quad q(dy_j) := f_j + I^2, \quad \ker(q) = \text{im}(\partial_2),$$

$$H_1(L_{R/Q}) = \text{coker}(\partial_2) \cong I/I^2.$$

Thus the linearized cotangent differential is the minimal conormal presentation. $\square$

Proposition 3.1.12 (Conormal sequence). *By Theorem 3.1.11,*

$$0 \longrightarrow \operatorname{im}(\partial_2) \longrightarrow \bigoplus_{x \in X_1} Rdx \xrightarrow{q} I/I^2 \longrightarrow 0, \quad q(dy_i) = f_i + I^2.$$

$$\begin{array}{ccccccc} 0 & \longrightarrow & \operatorname{im}(\partial_2) & \hookrightarrow & (\Omega_{A/Q} \otimes_A R)_1 & \twoheadrightarrow & \operatorname{coker}(\partial_2) \longrightarrow 0 \\ \parallel & & \downarrow \cong & & \downarrow \cong & & \downarrow \cong \\ 0 & \longrightarrow & \Omega_1^R(I/I^2) & \hookrightarrow & \bigoplus_{x \in X_1} Rdx & \xrightarrow{q} & I/I^2 \longrightarrow 0 \end{array}$$

Example 3.1.13 (Edge ideal of a triangle). Let

$$Q := k[x, y, z]_{(x, y, z)}, \quad I := (xy, xz, yz), \quad R := Q/I.$$

The minimal model begins with

$$\begin{aligned} X_1 &:= \{T_{1,1}, T_{1,2}, T_{1,3}\}, & \partial(T_{1,1}) &= xy, & \partial(T_{1,2}) &= xz, & \partial(T_{1,3}) &= yz, \\ X_2 &:= \{T_{2,1}, T_{2,2}\}, & \partial(T_{2,1}) &= z \cdot T_{1,1} - y \cdot T_{1,2}, & \partial(T_{2,2}) &= z \cdot T_{1,1} - x \cdot T_{1,3}, \\ X_3 &:= \{T_{3,1}, T_{3,2}, T_{3,3}\}, & \partial(T_{3,1}) &= T_{1,1} \cdot T_{1,2} + x \cdot T_{2,1}, \\ & & \partial(T_{3,2}) &= T_{1,2} \cdot T_{1,3} - z \cdot T_{2,1} + z \cdot T_{2,2}, & \partial(T_{3,3}) &= T_{1,1} \cdot T_{1,3} + y \cdot T_{2,2}, \\ R^2 &\xrightarrow{\begin{bmatrix} z & z \\ -y & 0 \\ 0 & -x \end{bmatrix}} R^3 \longrightarrow 0 \xrightarrow{3.1.11} I/I^2 \cong \frac{R^3}{R \begin{bmatrix} z \\ -y \\ 0 \end{bmatrix} + R \begin{bmatrix} z \\ 0 \\ -x \end{bmatrix}}. \end{aligned}$$

$$\cdots \longrightarrow L_3 = \bigoplus_{u \in X_3} Rdu \xrightarrow{\partial_3} L_2 = \bigoplus_{u \in X_2} Rdu \xrightarrow{\partial_2} L_1 = \bigoplus_{u \in X_1} Rdu \longrightarrow 0$$

$$\begin{array}{ccccc} RdT_{3,1} & \xrightarrow{x} & RdT_{2,1} & \xrightarrow{-y} & RdT_{1,2} \\ & \searrow -z & & \searrow z & \\ RdT_{3,2} & & & & RdT_{1,1} \\ & \searrow z & & \nearrow z & \\ RdT_{3,3} & \xrightarrow{y} & RdT_{2,2} & \xrightarrow{-x} & RdT_{1,3} \end{array}$$

3.2. Homotopy Lie.

Definition 3.2.1 (Graded Lie algebra). Let $L = \bigoplus_{n \geq 0} L^n$ be a graded k -vector space. A graded Lie algebra structure is

$$\begin{aligned} [-, -] : L^i \otimes_k L^j &\longrightarrow L^{i+j}, & (-)^{[2]} : L^{2i+1} &\longrightarrow L^{4i+2}, \\ [u, v] &= -(-1)^{|u||v|}[v, u], & [u, [v, w]] &= [[u, v], w] + (-1)^{|u||v|}[v, [u, w]], \\ [u, u] &= 2u^{[2]}, & (a \cdot u)^{[2]} &= a^2 \cdot u^{[2]}, & (u+v)^{[2]} &= u^{[2]} + [u, v] + v^{[2]}, & [u^{[2]}, w] &= [u, [u, w]] \end{aligned}$$

for homogeneous $u, v, w, a \in k$, and odd $|u| = |v|$ [1, §10].

Example 3.2.2 (Graded commutator). For a graded associative algebra B , set

$$B^{\text{Lie}} := B, \quad [a, b] := a \cdot b - (-1)^{|a||b|} b \cdot a, \quad u^{[2]} := u^2 \quad (|u| \text{ odd}).$$

Definition 3.2.3 (Universal enveloping algebra). For a graded Lie algebra L , set

$$J_L := (u \cdot v - (-1)^{|u||v|} v \cdot u - [u, v], w^2 - w^{[2]} : u, v, w \text{ homogeneous, } |w| \text{ odd}), \quad U(L) := T_k(L)/J_L.$$

For every graded associative algebra A ,

$$\varphi : L \longrightarrow A^{\text{Lie}} \implies \exists! \tilde{\varphi} : U(L) \longrightarrow A, \quad \tilde{\varphi} \iota = \varphi$$

[1, §10].

$$\begin{array}{ccc} & U(L) & \\ \iota \nearrow & & \searrow \exists! \tilde{\varphi} \\ L & \xrightarrow{\varphi} & A^{\text{Lie}} \end{array}$$

Theorem 3.2.4 (Poincaré–Birkhoff–Witt). For an ordered homogeneous basis $\{x_i\}$ of L ,

$$\mathcal{B}_L := \{x_{i_1} \cdots x_{i_r} : i_1 \leq \cdots \leq i_r, \text{ no odd } x_i \text{ repeated}\} \text{ is a } (k)\text{-basis of } U(L),$$

$$\text{gr } U(L) \cong \text{Sym}(L^{\text{even}}) \otimes_k \bigwedge_k (L^{\text{odd}})$$

[1, §10], [9].

$$\begin{array}{ccc} T(L) & \xrightarrow{q} & J = \left(\begin{array}{c} u \otimes v - (-1)^{|u||v|} v \otimes u - [u, v], \\ u \otimes u - u^{[2]} \end{array} \right) \\ \rho \downarrow & \searrow \text{gr}(q) & \\ \text{Sym}(L) & \xrightarrow[\text{PBW}]{\cong} & \text{gr } U(L) \end{array} \quad U(L) = T(L)/J,$$

Definition 3.2.5 (Homotopy Lie algebra). Set

$$\mathfrak{m}_1 := \mathfrak{m}_{A^{(1)}} = (X), \quad \text{ind}(A^{(1)}) := \mathfrak{m}_1 / \mathfrak{m}_1^2, \quad \pi^*(Q \rightarrow R) := (\Sigma \text{ind}(A^{(1)}))^{\vee}, \quad \pi^i(Q \rightarrow R) = (kX_{i-1})^{\vee}.$$

Minimality induces

$$\partial^{[2]} : \text{ind}(A^{(1)}) \longrightarrow \mathfrak{m}_1^2 / \mathfrak{m}_1^3, \quad \partial^{[2]}(\bar{x}) := \partial(x) \bmod \mathfrak{m}_1^3,$$

and, for $u \in \pi^i(Q \rightarrow R)$, $v \in \pi^j(Q \rightarrow R)$,

$$\langle u \otimes v, \xi \cdot \eta \rangle := \langle v, \xi \rangle \langle u, \eta \rangle + (-1)^{(i+1)(j+1)} \langle u, \xi \rangle \langle v, \eta \rangle, \quad \langle [u, v], \xi \rangle := (-1)^j \langle u \otimes v, \partial^{[2]}(\xi) \rangle.$$

The quadratic dual also defines $(-)^{[2]}$ [1, §10], [5].

$$\begin{array}{ccc} A^{(1)} = k[X] & \longleftarrow & \mathfrak{m}_{A^{(1)}} = (X) \xrightarrow{q} \text{ind}(A^{(1)}) = \mathfrak{m}_{A^{(1)}} / \mathfrak{m}_{A^{(1)}}^2 \\ & & \downarrow \Sigma \\ & & \Sigma \text{ind}(A^{(1)}) \xrightarrow{(-)^{\vee}} \pi^*(Q \rightarrow R) = (\Sigma \text{ind}(A^{(1)}))^{\vee} \end{array}$$

For a minimal acyclic closure $R\langle Y \rangle \xrightarrow{\sim} k$, set

$$\pi^*(R) := (kY)^{\vee}, \quad \pi^i(R) := (kY_i)^{\vee} \quad (i \geq 1),$$

with bracket and square dual to the quadratic closure differential.

Theorem 3.2.6 (Ext as an enveloping algebra). *There are canonical graded-algebra isomorphisms*

$$U(\pi^*(Q \rightarrow R)) \cong \text{Ext}_{A(1)}^*(k, k), \quad U(\pi^*(R)) \cong \text{Ext}_R^*(k, k),$$

$$Q \text{ regular} \implies \pi^i(Q \rightarrow R) \cong \pi^i(R) \quad (i \geq 2)$$

[1, §10], [12].

$$\begin{array}{ccc} Q[X] & \xrightarrow{\cong} & R \\ \text{ind}, \Sigma, (-)^\vee \downarrow & & \searrow \text{Yoneda} \\ \pi^*(R) & \xrightarrow[U(-)]{\quad} U(\pi^*(R)) & \xrightarrow[\cong]{} \text{Ext}_R^*(k, k) \end{array}$$

Theorem 3.2.7 (Martsinkovsky's criterion). *For a nonzero finitely generated R -module M ,*

$$\Omega_n^R(k) \twoheadrightarrow M, \quad \text{pd}_R(M) < \infty \quad (n \geq 0) \implies R \text{ regular}.$$

Proof. The nonzero syzygy quotient makes eventual Ext-vanishing detect every positive PBW generator.

$$\Omega_n^R(k) \twoheadrightarrow M \neq 0, \quad \text{Ext}_R^i(M, k) = 0 \quad (i \gg 0) \xrightarrow{3.2.4, 3.2.6, [1, \S 8]} \pi^i(R) = 0 \quad (i \geq 2),$$

$$\pi^i(R) = 0 \quad (i \geq 2) \xrightarrow{2.3.11, 2.3.12} \varepsilon_i(R) = 0 \quad (i \geq 2) \iff R \text{ regular}.$$

Thus the finite-projective-dimension quotient forces the singular homotopy classes to vanish. $\square$

3.3. Radical Detection. Throughout this subsection, assume

$$\text{pd}_Q(R) < \infty.$$

Definition 3.3.1 (Radical). For a graded ideal $\mathfrak{a} \triangleleft L$, set

$$\mathfrak{a}^{[0]} := \mathfrak{a}, \quad \mathfrak{a}^{[n+1]} := [\mathfrak{a}^{[n]}, \mathfrak{a}^{[n]}], \quad \mathfrak{a} \text{ solvable} \iff \mathfrak{a}^{[N]} = 0 \text{ for some } N,$$

$$\text{rad}(L) := \sum_{\mathfrak{a} \triangleleft L, \mathfrak{a} \text{ solvable}} \mathfrak{a}, \quad z \text{ radical} \iff z \in \text{rad}(L)$$

[1, §10], [5].

Proposition 3.3.2 (Eventual centrality). *For $z \in \pi^*(Q \rightarrow R)$,*

$$z \in \text{rad}(\pi^*(Q \rightarrow R)) \iff [z, \pi^{>n}(Q \rightarrow R)] = 0 \text{ for some } n$$

[5].

Theorem 3.3.3 (Avramov–Halperin).

$$Q \rightarrow R \text{ a complete-intersection homomorphism} \iff \pi^2(Q \rightarrow R) \subseteq \text{rad}(\pi^*(Q \rightarrow R))$$

[12], [5].

[Embedded deformations] Fix

$$Q \twoheadrightarrow S = Q/J \twoheadrightarrow R = Q/I, \quad J \subseteq I, \quad I = J + (f), \quad f \in \mathfrak{m}_S^2, \quad f \text{ (S)-regular}.$$

Then

$$I/I^2 \cong R\bar{f} \oplus C \implies f^\vee \in Z(\pi^*(Q \rightarrow R))_2, \quad [f^\vee, \pi^*(Q \rightarrow R)] = 0$$

[1, 5].

$$\begin{array}{ccccc}
Q & \xrightarrow{\quad} & S = Q/J & \xrightarrow{\quad \iota \quad} & R \cong S/(f) \\
& & J \subseteq I & & I = J + (f) \\
& \nearrow \text{produces} & & \searrow f \mapsto f^\vee & \\
I/I^2 \cong R\bar{f} \oplus C & & & & f \in \mathfrak{m}_S^2 \text{ is } S\text{-regular} \\
& & & & f^\vee \in Z(\pi^*(Q \rightarrow R))_2
\end{array}$$

Definition 3.3.4 (Differentiation along a homotopy class). For $z \in \pi^i(Q \rightarrow R) = (kX_{i-1})^\vee$, choose

$$\tilde{z}: QX_{i-1} \longrightarrow Q, \quad \tilde{z} \otimes_Q k = z,$$

and denote by

$$D_z := \frac{d}{dz} : A \longrightarrow A, \quad |D_z| = 1 - i, \quad D_z(x) := \begin{cases} \tilde{z}(x), & x \in X_{i-1}, \\ 0, & x \in X_j, j \neq i-1, \end{cases}$$

the unique Q -derivation with these values. Set

$$\theta_z := [\partial, D_z], \quad |\theta_z| = -i, \quad [\partial, \theta_z] = 0, \quad \theta_z(A) \subseteq \mathfrak{m}_A.$$

For $|\theta| \leq 0$, set

$$\theta^{(n)} : A^{(n)} \longrightarrow A^{(n)}, \quad \theta(\mathfrak{n}, X_{<n}) \subseteq (\mathfrak{n}, X_{<n}).$$

The class of θ_z is independent of $\tilde{z}$ [5].

Example 3.3.5 (Quadratic coefficient). For $z = y^\vee$ and

$$\partial(x) = y \cdot w + \text{terms not involving } y, \quad D_z(\partial x) = w + \dots, \quad \partial(D_z x) = 0 \xrightarrow{3.3.4} \theta_z(x) = w + \dots.$$

Thus θ_z differentiates the quadratic part of the minimal-model differential along z .

Proposition 3.3.6 (Adjoint action). For chain derivations $\theta, \theta' : A \rightarrow \mathfrak{m}_A$ of nonpositive degree,

$$\theta = \theta_z \implies \bar{\theta}^\vee = -\text{ad}(z) = -[z, -], \quad [\theta] = [\theta'] \implies \bar{\theta}^\vee = \bar{\theta}'^\vee.$$

$$\begin{array}{ccc}
(kX)^\vee & \xrightarrow{\bar{\theta}_z^\vee} & (kX)^\vee \\
\downarrow \simeq & & \downarrow \simeq \\
\pi^*(R) & \xrightarrow{-\text{ad}(z)} & \pi^*(R)
\end{array}$$

Proof. The quadratic differential controls both the bracket and the linear part of θ_z .

$$\partial(x) \equiv \sum_{u \leq v} c_{u,v}^x u \cdot v \pmod{\mathfrak{n}A + (X)^3}, \quad \theta_z(x) = [\partial, D_z](x) \equiv \sum_u c_{u,z}^x u \pmod{\mathfrak{m}_A^2},$$

$$\theta_z(x) \equiv \sum_u c_{u,z}^x u \pmod{\mathfrak{m}_A^2} \xrightarrow{3.2.5, 3.3.4} \bar{\theta}_z^\vee = -\text{ad}(z),$$

$$\theta - \theta' = [\partial, \sigma] \xrightarrow{3.1.7} (\theta - \theta')(\mathfrak{m}_A) \subseteq \mathfrak{m}_A^2 \implies \bar{\theta}^\vee = \bar{\theta}'^\vee.$$

Hence the action on dual indecomposables depends only on the derivation cohomology class [5]. $\square$

Corollary 3.3.7 (Vanishing on a cover). If $[\theta] = [\theta_z]$ and $\theta^{(n)} = 0$ for some n , then z is radical.

Proof. The induced map on the higher indecomposables is the restricted adjoint action.

$$[\theta] = [\theta_z], \quad \theta^{(n)} = 0 \xrightarrow{3.3.6} [z, \pi^{>n}(Q \rightarrow R)] = 0 \xrightarrow{3.3.2} z \in \text{rad}(\pi^*(Q \rightarrow R)).$$

Thus vanishing on one connected cover detects a radical class. $\square$

Lemma 3.3.8 (Core factorization). *For $z \in \pi^2(Q \rightarrow R)$, set*

$$\bar{\theta}_z : A \xrightarrow{\theta_z} \mathfrak{m}_A \twoheadrightarrow \mathfrak{m}.$$

Assume

$$\begin{aligned} \bar{\theta}_z &= \alpha \circ \beta, & \beta : A &\longrightarrow M \text{ chain derivation,} & \alpha &\in \text{Hom}_R(M, \mathfrak{m}), \\ & & M &\text{finitely generated,} & \text{pd}_R(M) &< \infty. \end{aligned}$$

Then z is radical.

$$\begin{array}{ccccc} \Omega_{A/Q} & \xrightarrow{\tilde{\beta}} & P & \xrightarrow{\tilde{\alpha}} & \mathfrak{m}_A \\ \omega \mapsto 1 \otimes \omega \downarrow & & \downarrow \simeq & & \downarrow \\ R \otimes_A \Omega_{A/Q} & \xrightarrow{\beta} & M & \xrightarrow{\alpha} & \mathfrak{m}_R \end{array}$$

Proof. A finite semifree resolution lifts the factorization, and its finitely many coefficients vanish on a high connected cover.

$$P \xrightarrow{\simeq} M, \quad P \text{ finite semifree over } A \xrightarrow{3.1.8, 2.2.7, [5]} \theta_z \simeq \theta := \tilde{\alpha} \circ \tilde{\beta} \circ d,$$

$$\text{Hom}_A(P, \mathfrak{m}_A) \cong \mathfrak{m}_A \otimes_A \text{Hom}_A(P, A), \quad \tilde{\alpha} = \sum_{j=1}^s a_j \cdot \sigma_j, \quad a_j \in \mathfrak{m}_A, \quad \sigma_j : P \rightarrow A,$$

$$n > \max\{|x| : x \text{ occurs in some } a_j\} \implies a_j \equiv 0 \pmod{(n, X_{<n})} \implies \theta^{(n)} = 0,$$

$$\theta_z \simeq \theta, \quad \theta^{(n)} = 0 \xrightarrow{3.3.7} z \in \text{rad}(\pi^*(Q \rightarrow R)).$$

Hence every finite-projective-dimension factorization annihilates the adjoint action above a finite degree. $\square$

Theorem 3.3.9 (Degree-two factorization). *There are natural isomorphisms*

$$\pi^2(Q \rightarrow R) \cong \text{Hom}_Q(QX_1, k) \cong \text{Hom}_Q(I, k) \cong \text{Hom}_R(I/I^2, k).$$

$$\pi^2(Q \rightarrow R) \xrightarrow{\cong} \text{Hom}_Q(QX_1, k) \xrightarrow{\cong} \text{Hom}_Q(I, k) \xrightarrow{\cong} \text{Hom}_R(I/I^2, k)$$

For $z \leftrightarrow \lambda_z : I/I^2 \rightarrow k$, assume

$$\lambda_z = \nu \circ \gamma, \quad I/I^2 \xrightarrow{\gamma} N \xrightarrow{\nu} k, \quad N \text{ finitely generated,} \quad \text{pd}_R(N) < \infty.$$

Then z is radical.

$$\begin{array}{ccc} R \otimes_A \Omega_{A/Q} & \xrightarrow{\beta} & M \\ & \searrow \theta = \alpha \circ \beta & \downarrow \alpha \\ & & R \end{array} \quad \text{pd}_R(M) < \infty$$

Proof. The degree-one variables identify conormal functionals, and the conormal presentation lifts their factorizations.

$$\begin{aligned} QX_1 \otimes_Q k &\cong I/\mathfrak{n}I, & I^2 &\subseteq \mathfrak{n}I, & (I/I^2) \otimes_R k &\cong I/\mathfrak{n}I, \\ \pi^2(Q \rightarrow R) &\cong \text{Hom}_k(I/\mathfrak{n}I, k) \cong \text{Hom}_Q(I, k) \cong \text{Hom}_R(I/I^2, k), \\ 0 &\longrightarrow M \longrightarrow F \longrightarrow N \longrightarrow 0, & F &\text{finite free,} & M &\subseteq \mathfrak{m}F, & \text{pd}_R(M) &< \infty, \end{aligned}$$

$$\lambda_z = \nu \circ \gamma \xrightarrow{3.1.11, [5]} \bar{\theta}_z = \left(A \xrightarrow{\beta} M \xrightarrow{\alpha} \mathfrak{m} \right) \xrightarrow{3.3.8} z \text{ radical.}$$

Thus finite conormal factorization produces the core derivation factorization. $\square$

Lemma 3.3.10 (Regular elements from finite projective dimension). *For a Noetherian local ring $(S, \mathfrak{m}_S)$,*

$$0 \neq J \subseteq S, \quad \text{pd}_S(J) < \infty \implies J \text{ contains an } (S)\text{-regular element.}$$

Proof. Localization at the associated primes converts a finite resolution of J into a constant-rank calculation.

$$\begin{aligned} 0 \longrightarrow F_d \longrightarrow \cdots \longrightarrow F_0 \longrightarrow J \longrightarrow 0, \quad \mathfrak{p} \in \text{Ass}(S), \quad \text{depth}(S_{\mathfrak{p}}) = 0, \\ \text{depth}(S_{\mathfrak{p}}) = 0 \xrightarrow{[6, \S 1.3]} J_{\mathfrak{p}} \text{ free}, \quad r := \text{rank}_{S_{\mathfrak{p}}}(J_{\mathfrak{p}}) = \sum_{i=0}^d (-1)^i \text{rank}_S(F_i), \\ \mathfrak{q} \in \text{Ass}(J) \subseteq \text{Ass}(S), \quad J_{\mathfrak{q}} \neq 0 \implies r > 0, \quad J_{\mathfrak{p}} \hookrightarrow S_{\mathfrak{p}} \implies r = 1, \quad J \not\subseteq \mathfrak{p}, \\ J \not\subseteq \mathfrak{p} \quad (\mathfrak{p} \in \text{Ass}(S)) \xrightarrow{[10, \S 1]} J \not\subseteq \bigcup_{\mathfrak{p} \in \text{Ass}(S)} \mathfrak{p} \xrightarrow{[10, \S 6]} J \text{ contains a nonzerodivisor.} \end{aligned}$$

Hence prime avoidance supplies the required regular element. $\square$

Theorem 3.3.11 (Ferrand–Vasconcelos). *For a surjective local homomorphism $Q \twoheadrightarrow R = Q/I$ with $\text{pd}_Q(R) < \infty$,*

$$I/I^2 \text{ free over } R \iff I \text{ generated by a } Q\text{-regular sequence}$$

[5].

Theorem 3.3.12 (Briggs). *Assume*

$$\alpha : I/I^2 \longrightarrow N, \quad N \text{ finitely generated}, \quad \text{pd}_R(N) < \infty, \quad \alpha \otimes_R k \text{ injective.}$$

Then I is generated by a Q -regular sequence [5].

Proof. Injectivity on the closed fibre makes every conormal functional factor through α .

$$\begin{aligned} \bar{\alpha} := \alpha \otimes_R k : (I/I^2) \otimes_R k \hookrightarrow N \otimes_R k \implies \bar{\alpha}^\vee : \text{Hom}_R(N, k) \twoheadrightarrow \text{Hom}_R(I/I^2, k), \\ \lambda : I/I^2 \longrightarrow k \text{ arbitrary} \implies \lambda \text{ factors through } N \xrightarrow{3.3.9} \pi^2(Q \rightarrow R) \subseteq \text{rad}(\pi^*(Q \rightarrow R)), \\ \pi^2(Q \rightarrow R) \subseteq \text{rad}(\pi^*(Q \rightarrow R)) \xrightarrow{3.3.3} Q \rightarrow R \text{ complete intersection} \xleftrightarrow{1.2.3} I \text{ generated by a } Q\text{-regular sequence.} \end{aligned}$$

Thus the fibrewise injection forces the complete-intersection condition. $\square$

Corollary 3.3.13 (Vasconcelos' conjecture).

$$\text{pd}_Q(R) < \infty, \quad \text{pd}_R(I/I^2) < \infty \implies I \text{ generated by a } Q\text{-regular sequence.}$$

Proof. Use the conormal module itself as the finite-projective-dimension target.

$$N := I/I^2, \quad \alpha := \text{id}_{I/I^2}, \quad \alpha \otimes_R k \text{ injective} \xrightarrow{3.3.12} I \text{ generated by a } Q\text{-regular sequence.}$$

Thus Theorem 3.3.12 applies directly. $\square$

Corollary 3.3.14 (Regular ambient ring). *For Q regular local and $R = Q/I$,*

$$R \text{ complete intersection} \iff \text{pd}_R(I/I^2) < \infty.$$

Proof. Regular sequences give free conormal modules, while regularity of Q supplies the ambient finite-projective-dimension hypothesis.

$$\begin{aligned} R \text{ complete intersection} \xrightarrow{1.2.3, [6, \S 2.3]} I/I^2 \text{ free over } R \implies \text{pd}_R(I/I^2) < \infty, \\ Q \text{ regular} \xrightarrow{1.2} \text{pd}_Q(R) < \infty, \quad \text{pd}_R(I/I^2) < \infty \xrightarrow{3.3.13} I \text{ (} Q\text{)-regular} \xrightarrow{1.2.3} R \text{ complete intersection.} \end{aligned}$$

Hence finite conormal projective dimension characterizes complete intersections over a regular local ring. $\square$

4. HYPERSURFACE PERIODICITY AND SUPPORT VARIETIES

Derived localization identifies Ext, higher homotopies produce hypersurface periodicity, and their multivariable form records complete-intersection support [14, 7, 1, 2, 11]. Fix

$$(Q, \mathfrak{n}, k) \text{ regular local, } \quad Q \twoheadrightarrow (R, \mathfrak{m}, k) = Q/I, \quad I = (f_1, \dots, f_c) \subseteq \mathfrak{n}^2 \text{ minimal,}$$

$$E := K^Q(f_1, \dots, f_c) = Q\langle e_1, \dots, e_c \mid \partial(e_i) = f_i \rangle, \quad S := k[\chi_1, \dots, \chi_c], \quad |\chi_i| = 2.$$

4.1. Derived Categories.

Definition 4.1.1 (Homotopy, shift, and cone). For chain maps $f, g : X \rightarrow Y$, set [16, §1.5]

$$f \simeq g \iff f - g = \partial_Y h + h \partial_X \quad \text{for some } h : X \rightarrow Y, |h| = 1,$$

$$(\Sigma^n X)_i = X_{i-n}, \quad \partial_{\Sigma^n X} = (-1)^n \partial_X, \quad \text{cone}(f)_i = Y_i \oplus X_{i-1}, \quad \text{Cone}(f) := \text{cone}(f),$$

$$\partial_{\text{cone}(f)}(y, x) = (\partial_Y(y) + f(x), -\partial_X(x)).$$

$$\text{Cone}(f) = Y \oplus \Sigma X$$

$$X \xrightarrow{f} Y \xrightarrow{j} \partial_{\text{Cone}(f)} = \begin{bmatrix} \partial_Y & f \\ 0 & -\partial_X \end{bmatrix} \xrightarrow{q} \Sigma X$$

Proposition 4.1.2 (Short exact triangles). For $0 \rightarrow X \xrightarrow{i} Y \xrightarrow{p} Z \rightarrow 0$, the map ρ below is a quasi-isomorphism.

$$\begin{array}{ccccccc} 0 & \longrightarrow & X & \xrightarrow{i} & Y & \xrightarrow{p} & Z \longrightarrow 0 \\ & & \parallel & & \parallel & & \uparrow \rho \simeq \\ & & X & \xrightarrow{i} & Y & \xrightarrow{j} & \text{Cone}(i) \xrightarrow{q} \Sigma X \end{array}$$

Proof. The cone contracts precisely along the kernel of the quotient map.

$$\rho : \text{cone}(i) \longrightarrow Z, \quad \rho(y, x) = p(y), \quad \ker(\rho) \cong \text{cone}(\text{id}_X),$$

$$\text{cone}(\text{id}_X) \simeq 0 \implies \rho \text{ a quasi-isomorphism} \xrightarrow{4.1.1} (X \rightarrow Y \rightarrow Z \rightarrow \Sigma X) \cong (X \rightarrow Y \rightarrow \text{cone}(i) \rightarrow \Sigma X).$$

Thus the short exact sequence realizes the canonical cone triangle. $\square$

Definition 4.1.3 (Derived category). Set [16, §10]

$$K(R) := \text{Ch}(R) / \simeq, \quad D(R) := K(R)[\text{quasi-isomorphisms}^{-1}].$$

$$s \text{ quasi-isomorphism} \iff H_i(s) \text{ isomorphism} \quad (i \in \mathbb{Z}).$$

$$\text{Hom}_{D(R)}(X, Y) \ni [X \xleftarrow{\simeq} X' \longrightarrow Y].$$

$$\begin{array}{ccccccc} \text{Ch}(R) \ni X_0 & \xrightarrow{\simeq} & X_1 & \xleftarrow{\simeq} & X_2 & \xrightarrow{\simeq} & X_3 \\ \downarrow \mathcal{L} & & & & & & \downarrow \mathcal{L} \\ D(R) = \text{Ch}(R)[\text{qis}^{-1}] \ni \mathcal{L}X_0 & \xrightarrow{\simeq} & & & & & \mathcal{L}X_3 \end{array}$$

Proposition 4.1.4 (Acyclic factorizations). For $H(X) = 0$, every morphism in $D(R)$ factoring through X is zero.

$$\begin{array}{ccc}
M & \xrightarrow{f=h \circ g} & N \\
& \searrow g & \nearrow h \\
& X & \\
& H(X) = 0 & \\
& \downarrow \simeq & \\
& 0 &
\end{array}$$

Proof. Acyclic objects vanish after localization.

$$H(X) = 0 \implies 0 \xrightarrow{\simeq} X \xrightarrow{4.1.3} X \cong 0 \text{ in } D(R) \implies h \circ g = 0.$$

Thus every such factorization is zero. $\square$

Theorem 4.1.5 (Derived description of Ext). *For R -modules M, N , free resolutions $F_M \rightarrow M$, $F_N \rightarrow N$, and $n \geq 0$,*

$$\mathrm{Hom}_{D(R)}(M, \Sigma^n N) \cong H_0(\mathrm{Hom}_R(F_M, \Sigma^n F_N)) \cong \mathrm{Ext}_R^n(M, N)$$

[16, §10].

$$\mathrm{Hom}_{D(R)}(M, \Sigma^n N) \xrightarrow{\simeq} H_0 \mathrm{Hom}_R(F_M, \Sigma^n F_N) \xrightarrow{\simeq} H_{-n} \mathrm{Hom}_R(F_M, F_N) \xrightarrow{\simeq} \mathrm{Ext}_R^n(M, N)$$

Proof. Free resolutions compute both the derived morphism and classical Ext groups.

$$\begin{aligned}
F_M &\xrightarrow{\simeq} M, & F_N &\xrightarrow{\simeq} N \xrightarrow{4.1.3} \mathrm{Hom}_{D(R)}(M, \Sigma^n N) \cong \mathrm{Hom}_{K(R)}(F_M, \Sigma^n F_N), \\
& & & \mathrm{Hom}_{K(R)}(F_M, \Sigma^n F_N) = H_0(\mathrm{Hom}_R(F_M, \Sigma^n F_N)), \\
& & & F_M \text{ bounded below and free} \xrightarrow{[16, \S 10]} \mathrm{Hom}_R(F_M, F_N) \xrightarrow{\simeq} \mathrm{Hom}_R(F_M, N), \\
& & & H_{-n}(\mathrm{Hom}_R(F_M, F_N)) \cong H^n(\mathrm{Hom}_R(F_M, N)) = \mathrm{Ext}_R^n(M, N).
\end{aligned}$$

Hence derived morphisms recover the usual Ext groups. $\square$

Theorem 4.1.6 (Long exact sequences). *For a distinguished triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} \Sigma X$ and $M \in D(R)$, the functor $\mathrm{Hom}_{D(R)}(M, -)$ gives the sequence below; for modules it is the usual Ext sequence [16, §1.5, §10], [13].*

$$\begin{array}{ccccccc}
\mathrm{Hom}_{D(R)}(M, \Sigma^n X) & \xrightarrow{(\Sigma^n u)_*} & \mathrm{Hom}_{D(R)}(M, \Sigma^n Y) & \xrightarrow{(\Sigma^n v)_*} & \mathrm{Hom}_{D(R)}(M, \Sigma^n Z) & & \\
& & & \nwarrow (\Sigma^n w)_* & & & \\
\mathrm{Hom}_{D(R)}(M, \Sigma^{n+1} X) & \xrightarrow{(\Sigma^{n+1} u)_*} & \mathrm{Hom}_{D(R)}(M, \Sigma^{n+1} Y) & \xrightarrow{(\Sigma^{n+1} v)_*} & \mathrm{Hom}_{D(R)}(M, \Sigma^{n+1} Z) & \longrightarrow & \dots
\end{array}$$

Definition 4.1.7 (Semifree DG module). For a DG A -module F , set

$$F \text{ semifree} \iff \left[0 = F(-1) \subseteq F(0) \subseteq F(1) \subseteq \dots, \quad F = \bigcup_{i \geq 0} F(i), \quad F(i)/F(i-1) \cong \bigoplus_{\lambda \in \Lambda_i} \Sigma^{d_\lambda} A \right]$$

[1, §1].

Theorem 4.1.8 (Semifree lifting). *Every DG A -module admits a surjective semifree resolution, and every map lifts uniquely up to homotopy [1, §1]. For $f : M \rightarrow N$, choose*

$$F_M, F_N \text{ semifree}, \quad F_M \twoheadrightarrow M, \quad F_N \twoheadrightarrow N \text{ surjective quasi-isomorphisms.}$$

$$\begin{array}{ccc}
F_M & \xrightarrow{\tilde{f}} & F_N \\
\downarrow \simeq & & \downarrow \simeq \\
M & \xrightarrow{f} & N
\end{array}$$

Corollary 4.1.9 (Lifted quasi-isomorphisms). *A lift between semifree resolutions of a quasi-isomorphism is a homotopy equivalence.*

Proof. Semifree lifting and semiprojectivity supply a homotopy inverse.

$$f : M \xrightarrow{\simeq} N, \quad \tilde{f} : F_M \rightarrow F_N, \quad F_M, F_N \text{ semifree},$$

$$\text{Hom}_A(F_N, F_M) \xrightarrow{\tilde{f} \circ -} \text{Hom}_A(F_N, F_N) \text{ a quasi-isomorphism} \implies \exists g : F_N \rightarrow F_M, \quad \tilde{f} \circ g \simeq \text{id}_{F_N},$$

$$\tilde{f} \circ (g \circ \tilde{f} - \text{id}_{F_M}) \simeq 0, \quad \text{Hom}_A(F_M, F_M) \xrightarrow{\tilde{f} \circ -} \text{Hom}_A(F_M, F_N) \text{ a quasi-isomorphism} \implies g \circ \tilde{f} \simeq \text{id}_{F_M}.$$

Thus $\tilde{f}$ is a homotopy equivalence. $\square$

4.2. Higher Homotopies and Periodicity. Throughout

$$c = 1, \quad f := f_1 \neq 0, \quad I = (f), \quad R = Q/(f).$$

Definition 4.2.1 (Higher homotopies). For a finitely generated R -module M and a finite free Q -resolution $F \rightarrow M$, a system of higher homotopies for f consists of [14, 7]

$$\sigma_i : F \rightarrow F \text{ } Q\text{-linear} \quad (i \geq 0), \quad |\sigma_i| = 2i - 1, \quad \sigma_0 = \partial_F,$$

$$\sum_{i+j=n} \sigma_i \sigma_j = \begin{cases} f \text{id}_F, & n = 1, \\ 0, & n \neq 1 \end{cases} \quad (n \geq 0)$$

$$\begin{array}{ccccccc}
\Sigma^{-1}F & & \Sigma F & & \Sigma^3 F & & \Sigma^5 F & \dots \\
& \searrow \sigma_0 = \partial_F & & \searrow \sigma_1 & \downarrow \sigma_2 & \swarrow \sigma_3 & & \\
& & & & F & & &
\end{array}$$

$$|\sigma_i| = 2i - 1$$

$$\sum_{i+j=1} \sigma_i \sigma_j = f \text{id}_F$$

$$\sum_{i+j=n} \sigma_i \sigma_j = 0 \quad (n \geq 0, n \neq 1)$$

Example 4.2.2 (An explicit higher homotopy). Let

$$Q := k[[x, y]], \quad f := x^2, \quad M := Q/(x^2, xy).$$

For the resolution in Example 3.1.5, set

$$\sigma_1(1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \sigma_1 \begin{bmatrix} a \\ b \end{bmatrix} = b \cdot x, \quad \sigma_1(F_2) = 0,$$

$$\partial_F \sigma_1 + \sigma_1 \partial_F = x^2 \text{id}_F, \quad \sigma_1^2 = 0 \xrightarrow{4.2.1} (\sigma_0, \sigma_1, 0, \dots) \text{ a system of higher homotopies.}$$

$$\begin{array}{ccccccc}
0 & \longrightarrow & Q & \begin{array}{c} \xrightarrow{\partial_2 = \begin{bmatrix} -y \\ x \end{bmatrix}} \\ \xleftarrow{(\sigma_1)_1 = \begin{bmatrix} 0 & x \end{bmatrix}} \end{array} & Q^2 & \begin{array}{c} \xleftarrow{\partial_1 = \begin{bmatrix} x^2 & xy \end{bmatrix}} \\ \xrightarrow{(\sigma_1)_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}} \end{array} & Q & \longrightarrow & 0
\end{array}$$

$$\partial \sigma_1 + \sigma_1 \partial = x^2 \text{id}_F$$

Definition 4.2.3 (Shamash complex). For a system σ as in Definition 4.2.1, set [14], [1, §3]

$$\Gamma_R(\chi) := \bigoplus_{i \geq 0} R\chi^{(i)}, \quad \chi^{(i)} \text{ divided-power basis,} \quad |\chi^{(i)}| = 2i, \quad \chi^{(0)} = 1, \quad \chi^{(-1)} := 0,$$

$$\chi^{(i)} \chi^{(j)} = \binom{i+j}{i} \chi^{(i+j)} \quad (i, j \geq 0).$$

$$\text{Sh}(F, \sigma)_n := \bigoplus_{i \geq 0} R\chi^{(i)} \otimes_Q F_{n-2i}, \quad \partial_{\text{Sh}}(\chi^{(i)} \otimes u) := \sum_{j=0}^i \chi^{(i-j)} \otimes \sigma_j(u).$$

Proposition 4.2.4 (Shamash differential). *The map ∂_{Sh} satisfies $\partial_{\text{Sh}}^2 = 0$.*

$$\begin{array}{ccc}
& \sigma_2 & \\
\begin{array}{c} \downarrow \\ R\chi^{(0)} \otimes F_n \xrightarrow{\sigma_0} R\chi^{(0)} \otimes F_{n-1} \\ \uparrow \\ R\chi^{(1)} \otimes F_{n-2} \xrightarrow{\sigma_0} R\chi^{(1)} \otimes F_{n-3} \\ \uparrow \\ R\chi^{(2)} \otimes F_{n-4} \xrightarrow{\sigma_0} R\chi^{(2)} \otimes F_{n-5} \end{array} & \begin{array}{c} \xleftarrow{\sigma_1} \\ \xleftarrow{\sigma_1} \\ \xleftarrow{\sigma_1} \end{array} & \begin{array}{c} \downarrow \\ \downarrow \\ \downarrow \end{array}
\end{array}$$

$\vdots$

Proof. The higher-homotopy identities collapse the square to multiplication by f .

$$\partial_{\text{Sh}}^2(\chi^{(i)} \otimes u) \xLeftrightarrow{4.2.3} \sum_{m=0}^i \chi^{(i-m)} \otimes \left(\sum_{j+\ell=m} \sigma_j \sigma_\ell \right)(u),$$

$$\sum_{m=0}^i \chi^{(i-m)} \otimes \left(\sum_{j+\ell=m} \sigma_j \sigma_\ell \right)(u) \xLeftrightarrow{4.2.1} \chi^{(i-1)} \otimes f \cdot u = 0 \quad \text{over } R = Q/(f).$$

Thus the Shamash formula defines a complex. □

Theorem 4.2.5 (Shamash). *For $F \rightarrow M$ and σ as in Definition 4.2.1 [14], [1, §3],*

$$\text{Sh}(F, \sigma) \xrightarrow{\simeq} M \quad \text{is a free } R\text{-resolution.}$$

For $\text{pd}_Q(M) = 2d$, write

$$\begin{array}{ccccccccc}
2d+4 & & 2d+3 & & 2d+2 & & 2d+1 & & 2d \\
R\chi^{(2)} \otimes F_{2d} & \xrightarrow{\sigma_0} & R\chi^{(2)} \otimes F_{2d-1} & \xrightarrow{\sigma_1} & R\chi^{(1)} \otimes F_{2d} & \xrightarrow{\sigma_0} & R\chi^{(1)} \otimes F_{2d-1} & \xrightarrow{\sigma_1} & R\chi^{(0)} \otimes F_{2d} \\
\oplus & & \oplus & & \oplus & & \oplus & & \oplus \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots \\
\oplus & & \oplus & & \oplus & & \oplus & & \oplus \\
R\chi^{(d+1)} \otimes F_2 & \xrightarrow{\sigma_1} & R\chi^{(d)} \otimes F_3 & \xrightarrow{\sigma_0} & R\chi^{(d)} \otimes F_2 & \xrightarrow{\sigma_1} & R\chi^{(d-1)} \otimes F_3 & \xrightarrow{\sigma_0} & R\chi^{(d-1)} \otimes F_2 \\
\oplus & & \oplus & & \oplus & & \oplus & & \oplus \\
R\chi^{(d+2)} \otimes F_0 & \xrightarrow{\sigma_1} & R\chi^{(d+1)} \otimes F_1 & \xrightarrow{\sigma_0} & R\chi^{(d+1)} \otimes F_0 & \xrightarrow{\sigma_1} & R\chi^{(d)} \otimes F_1 & \xrightarrow{\sigma_0} & R\chi^{(d)} \otimes F_0
\end{array}$$

δ^d (dashed arrows from $R\chi^{(2)} \otimes F_{2d}$ to $R\chi^{(d+1)} \otimes F_2$), δ^{γ} (dashed arrows from $R\chi^{(d+1)} \otimes F_2$ to $R\chi^{(d)} \otimes F_3$), δ^{γ} (dashed arrows from $R\chi^{(d)} \otimes F_3$ to $R\chi^{(d)} \otimes F_2$), δ^{γ} (dashed arrows from $R\chi^{(d)} \otimes F_2$ to $R\chi^{(d-1)} \otimes F_3$), δ^{γ} (dashed arrows from $R\chi^{(d-1)} \otimes F_3$ to $R\chi^{(d-1)} \otimes F_2$), δ^{γ} (dashed arrows from $R\chi^{(d-1)} \otimes F_2$ to $R\chi^{(d)} \otimes F_1$), δ^{γ} (dashed arrows from $R\chi^{(d)} \otimes F_1$ to $R\chi^{(d)} \otimes F_0$).

Definition 4.2.6 (Matrix factorization). For finite free Q -modules G_0, G_1 , a matrix factorization of f is

$$G_1 \xrightarrow{\varphi} G_0 \xrightarrow{\psi} G_1, \quad \varphi\psi = f \text{id}_{G_0}, \quad \psi\varphi = f \text{id}_{G_1}.$$

Reduction modulo f gives

$$\overline{\varphi} \overline{\psi} = \overline{\psi} \overline{\varphi} = 0$$

and hence a 2-periodic complex of finite free R -modules [7].

Theorem 4.2.7 (Eisenbud periodicity). For $\Omega_n^R(M)$ as in Definition 1.1.4, there exist $n_0 \geq 0$ and a matrix factorization (A, B) of f such that

$$\Omega_{n_0}^R(M) \cong \text{coker}(\overline{A}), \quad \Omega_{n_0+1}^R(M) \cong \text{coker}(\overline{B}), \quad \Omega_{n+2}^R(M) \cong \Omega_n^R(M) \quad (n \geq n_0)$$

[7], [1, §3].

$$\cdots \longrightarrow R^b \xrightarrow{\overline{B}} R^b \xrightarrow{\overline{A}} R^b \xrightarrow{\overline{B}} R^b \xrightarrow{\overline{A}} \cdots$$

Corollary 4.2.8 (Periodic invariants). For a finitely generated R -module M and $n \gg 0$,

$$\text{Tor}_{n+2}^R(M, k) \cong \text{Tor}_n^R(M, k), \quad \text{Ext}_R^{n+2}(M, k) \cong \text{Ext}_R^n(M, k), \quad \beta_{n+2}^R(M) = \beta_n^R(M).$$

$$\text{Tor}_n^R(M, k) \xrightarrow{\cong} \text{Tor}_{n+2}^R(M, k)$$

$$\text{Ext}_R^n(M, k) \xrightarrow{\cong} \text{Ext}_R^{n+2}(M, k)$$

Proof. Let $F \rightarrow M$ be minimal; tensoring or dualizing its periodic tail preserves two-step repetition.

$$F \xrightarrow{\cong} M \text{ minimal free} \xrightarrow{4.2.7, 4.2.6} F_{n+2} \cong F_n, \quad \partial_{n+2} \cong \partial_n \quad (n \gg 0),$$

$$F_{n+2} \cong F_n, \quad \partial_{n+2} \cong \partial_n \implies H_{n+2}(F \otimes_R k) \cong H_n(F \otimes_R k), \quad H^{n+2}(\text{Hom}_R(F, k)) \cong H^n(\text{Hom}_R(F, k)),$$

$$\text{Tor}_{n+2}^R(M, k) \cong \text{Tor}_n^R(M, k), \quad \text{Ext}_R^{n+2}(M, k) \cong \text{Ext}_R^n(M, k) \implies \beta_{n+2}^R(M) = \beta_n^R(M).$$

Thus Tor, Ext, and Betti numbers inherit the periodic tail. $\square$

4.3. Operators and Support Varieties. Throughout

$f_1, \dots, f_c$ a Q -regular sequence, $k = \bar{k}$.

Definition 4.3.1 (Multivariable higher homotopies). For a finitely generated R -module M , a finite free Q -resolution $F \rightarrow M$, and $\alpha \in \mathbb{N}^c$, set

$$|\alpha| := \alpha_1 + \dots + \alpha_c, \quad \mathbf{e}_1, \dots, \mathbf{e}_c \text{ the standard basis of } \mathbb{N}^c, \quad |\sigma_\alpha| := 2|\alpha| - 1.$$

A system of higher homotopies is

$$\begin{aligned} \{\sigma_\alpha : F \longrightarrow F \text{ } Q\text{-linear}\}_{\alpha \in \mathbb{N}^c}, \quad \sigma_0 = \partial_F, \quad \sigma_0 \sigma_{\mathbf{e}_i} + \sigma_{\mathbf{e}_i} \sigma_0 = f_i \text{id}_F, \\ \sum_{\alpha+\beta=\gamma} \sigma_\alpha \sigma_\beta = 0 \quad (|\gamma| \geq 2), \quad c = 1 \xrightarrow{4.2.1} \sigma_{(i)} = \sigma_i \quad (i \geq 0). \end{aligned}$$

Set

$$\begin{aligned} \Gamma_R(\chi) &:= \bigoplus_{\alpha \in \mathbb{N}^c} R\chi^{(\alpha)}, \quad |\chi^{(\alpha)}| = 2|\alpha|, \quad \chi^{(0)} = 1, \\ \chi^{(\alpha)} \chi^{(\beta)} &= \binom{\alpha+\beta}{\alpha} \chi^{(\alpha+\beta)}, \quad \text{Sh}(F, \sigma; \mathbf{f})_n := \bigoplus_{\alpha \in \mathbb{N}^c} R\chi^{(\alpha)} \otimes_Q F_{n-2|\alpha|}, \\ \partial_{\text{Sh}}(\chi^{(\alpha)} \otimes u) &:= \sum_{0 \leq \beta \leq \alpha} \chi^{(\alpha-\beta)} \otimes \sigma_\beta(u), \quad \partial_{\text{Sh}}^2 = 0, \quad \text{Sh}(F, \sigma; \mathbf{f}) \xrightarrow{\sim} M \end{aligned}$$

[7, 2].

$$\begin{array}{ccccc} R\chi_1^{(2)} \otimes_Q F & & R\chi_1^{(1)} \chi_2^{(1)} \otimes_Q F & & R\chi_2^{(2)} \otimes_Q F \\ & \searrow \sigma_{1,0} & \swarrow \sigma_{0,1} \quad \searrow \sigma_{1,0} & & \swarrow \sigma_{0,1} \\ & R\chi_1^{(1)} \otimes_Q F & & R\chi_2^{(1)} \otimes_Q F & \\ & \searrow \sigma_{1,0} & & \swarrow \sigma_{0,1} & \\ & R \otimes_Q F & & & \end{array}$$

Theorem 4.3.2 (Cohomology operators). For every finitely generated R -module M , the multivariable Shamash construction induces

$$\chi_i : \text{Ext}_R^n(M, k) \longrightarrow \text{Ext}_R^{n+2}(M, k), \quad \chi_i \chi_j = \chi_j \chi_i, \quad \text{Ext}_R^*(M, k) \text{ finitely generated over } S$$

[7, 2].

Definition 4.3.3 (Cohomological and rank support). For a finitely generated R -module M , set

$$\begin{aligned} \mathbb{A}_k^c &:= \text{Spec}(S), \quad \mathbb{A}_k^c(k) := \text{MaxSpec}(S) = k^c, \quad 0 := (\chi_1, \dots, \chi_c) \in \mathbb{A}_k^c(k), \\ V_E(M) &:= \text{Supp}_S(\text{Ext}_E^*(M, k)) \subseteq \mathbb{A}_k^c, \quad E \xrightarrow{\sim} R, \quad V_R(M) := V_E(M) \cap \mathbb{A}_k^c(k) \end{aligned}$$

[2].

$$S = k[\chi_1, \dots, \chi_c] \xrightarrow{\rho_M} \text{End}_k^*(\text{Ext}_E^*(M, k))$$

$$V_E(M) = \text{Supp}_S \text{Ext}_E^*(M, k) \hookrightarrow \text{Spec}(S) = \mathbb{A}_k^c$$

For $a = (a_1, \dots, a_c) \in k^c$, choose $\tilde{a}_i + \mathfrak{n} = a_i$ and set

$$f_a := \sum_{i=1}^c \tilde{a}_i \cdot f_i, \quad Q_a := Q/(f_a), \quad V_R^{\text{rank}}(M) := \{0\} \cup \{a \in k^c \setminus \{0\} : \text{pd}_{Q_a}(M) = \infty\}.$$

This depends only on $f_a + \mathfrak{n}I \in I/\mathfrak{n}I$ [2].

Theorem 4.3.4 (Rank detection and specialization). *For nonzero finitely generated R -modules M, N ,*

$$V_R(M) = V_R^{\text{rank}}(M), \quad V_R(M) = \{0\} \iff \text{pd}_R(M) < \infty, \quad V_R(k) = \mathbb{A}_k^c(k), \quad V_R(M \oplus N) = V_R(M) \cup V_R(N).$$

For a finite free Q -resolution $F \rightarrow M$, set

$$F_{\text{ev}} := \bigoplus_{i \geq 0} F_{2i}, \quad F_{\text{odd}} := \bigoplus_{i \geq 0} F_{2i+1}, \quad N_0 := F_{\text{ev}} \otimes_Q k, \quad N_1 := F_{\text{odd}} \otimes_Q k.$$

The higher homotopies determine

$$\begin{aligned} T &:= Q[t_1, \dots, t_c], \quad \tilde{A}(\mathbf{t}) : T \otimes_Q F_{\text{ev}} \longrightarrow T \otimes_Q F_{\text{odd}}, \quad \tilde{B}(\mathbf{t}) : T \otimes_Q F_{\text{odd}} \longrightarrow T \otimes_Q F_{\text{ev}}, \\ \tilde{A}(\mathbf{t})\tilde{B}(\mathbf{t}) &= \tilde{B}(\mathbf{t})\tilde{A}(\mathbf{t}) = \left(\sum_{i=1}^c t_i f_i \right) \text{id}, \\ T &\twoheadrightarrow S, \quad t_i \mapsto \chi_i, \quad A(\chi) := \overline{\tilde{A}(\mathbf{t})}, \quad B(\chi) := \overline{\tilde{B}(\mathbf{t})}, \quad A(\chi)B(\chi) = B(\chi)A(\chi) = 0, \\ a \in V_R(M) &\xleftrightarrow{4.2.7, 4.2.6} \left(N_0 \xrightarrow{A(a)} N_1 \xrightarrow{B(a)} N_0 \right) \text{ not exact} \end{aligned}$$

[7, 2].

$$\begin{aligned} \dots &\longrightarrow N_0 \xrightarrow{A(a)} N_1 \xrightarrow{B(a)} N_0 \xrightarrow{A(a)} N_1 \xrightarrow{B(a)} \dots \\ \dots &\longrightarrow S \otimes_k N_0 \xrightarrow{A(\chi)} S \otimes_k N_1 \xrightarrow{B(\chi)} S \otimes_k N_0 \xrightarrow{A(\chi)} S \otimes_k N_1 \xrightarrow{B(\chi)} \dots \end{aligned}$$

4.4. Koszul Detection. Throughout

$f_1, \dots, f_c$ not assumed Q -regular.

Definition 4.4.1 (Koszul support). For $M \in D^f(R)$, set [11]

$$\mathcal{V}_E(M) := \text{Supp}_S(\text{Ext}_E^*(M, k)) \subseteq \text{Spec}(S).$$

$$f_1, \dots, f_c \text{ } Q\text{-regular, } M \text{ an } R\text{-module} \implies \mathcal{V}_E(M) \cap \text{MaxSpec}(S) = V_R(M).$$

$$f_1, \dots, f_c \text{ } Q\text{-regular} \implies E = Q[X_1] = Q[X], \quad E \xrightarrow{\sim} R$$

Choose

$$\begin{array}{ccc} [R \longrightarrow Q[X]] & := & [Q[X] \xrightarrow{\sim} R]^{-1} \quad \text{in } D(Q). \\ Q & \twoheadrightarrow & R = Q/I \\ & \searrow & \downarrow \\ & & E = K^Q(f_1, \dots, f_c) \xrightarrow{\sim} Q[X] \xrightarrow{\sim} R \end{array}$$

Theorem 4.4.2 (Operators over the Koszul stage). *For $M \in D(E)$ with finitely generated total homology, the Yoneda action induces a natural graded S -module structure*

$$S \hookrightarrow \text{Ext}_E^*(k, k) \longrightarrow \text{End}_k^*(\text{Ext}_E^*(M, k)).$$

$$P_M \text{ semifree,} \quad P_M \twoheadrightarrow M \text{ a quasi-isomorphism,} \quad \text{Ext}_E^*(M, k) \cong H^*(\text{Hom}_E(P_M, k)).$$

$$M \in D^f(R) \xrightarrow{4.4.1, [2, 11]} \text{Ext}_E^*(M, k) \text{ finitely generated over } S.$$

Proposition 4.4.3 (Derivations and operators). *For a minimal model $Q[X] \xrightarrow{\sim} R$, $z \in \pi^2(Q \rightarrow R)$, and θ_z as in Definition 3.3.4, set*

$$\theta_z(E) = 0, \quad \chi_z := H^*(- \circ \theta_z) : \text{Ext}_E^*(R, k) \longrightarrow \text{Ext}_E^{*+2}(R, k).$$

$$\begin{array}{ccc} \text{Hom}_E(Q[X], k) & \xrightarrow{- \circ \theta_z} & \text{Hom}_E(Q[X], k) \\ \downarrow H^* & & \downarrow H^* \\ \text{Ext}_E^*(R, k) & \xrightarrow{\chi_z} & \text{Ext}_E^{*+2}(R, k) \end{array}$$

Centrality forces eventual nilpotence.

$$z \text{ central} \xrightarrow{3.3.2, [5, \text{Theorem 3.1}]} \chi_z^n = H^*(- \circ \theta_z^n) = 0 \quad (n \gg 0).$$

After taking H^0 on the left and H^{2n} on the right, the comparison is represented schematically below.

$$\begin{array}{ccc} \text{Hom}_E(Q[X], k) & \xrightarrow{- \circ \theta_z^n} & \text{Hom}_E(Q[X], k) \\ \downarrow \simeq & & \downarrow \simeq \\ \text{Hom}_{D(E)}(R, k) & \xrightarrow{\chi_z^n} & \text{Hom}_{D(E)}(R, \Sigma^{2n}k) \end{array}$$

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